Error-field induced homoclinic tangles in TCV

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Abstract

Non-axisymmetric magnetic perturbations in tokamaks can lead to chaotic magnetic field line behaviour and to the formation of magnetic tangles near the separatrix. In TCV, unusual emissivity patterns have been observed in the divertor region using the MANTIS diagnostic.

In this work, a possible link between these structures and error-field-induced magnetic tangles is investigated. Tomographic reconstructions of MANTIS measurements are compared with magnetic field-line simulations based on Poincaré maps, using axisymmetric equilibria from LIUQE and non-axisymmetric perturbations estimated from poloidal field coil displacements. A qualitative spatial correlation is observed between simulated turnstile lobes and the patterns measured by the MANTIS diagnostic.

These results support the interpretation of these patterns being signatures of turnstile transport induced by error-field-driven magnetic tangles in TCV.

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1 Introduction

Nuclear fusion is one of the most promising technologies for generating sustainable energy, due to its high density and low-carbon emissions. However, the creation of any new large-scale energy technology often comes with unintended consequences and rebound effects. If the prevailing profit-driven economic model does not change before the implementation of fusion energy, the availability of a new abundant energy source could stimulate an increase in industrial production. This would lead to even more material production and waste generation instead of reducing the overall environmental impact. These considerations underline the importance of examining fusion energy not as a singular solution for sustainable energy production, but as a potential tool within a broader framework aimed at achieving a globally sustainable society. Even with these considerations, its development could still lead to a significant reduction in CO2 emissions and provide cheaper energy for society.

The fusion process is based on the merging of two light nuclei into one heavier nuclei, which releases a considerable amount of energy. As this is the fuel that powers stars, one can imagine the amount of energy required to make fusion happen. It requires extreme temperature at a sufficiently high density, implying that atoms must be in a plasma state. Often mentioned as the "fourth" matter state, the plasma is similar to a really hot gas, but with the specificity of having a large amount of charged particles with still having a global neutral charge. Charged particles are sensitive to magnetic fields, they follow the magnetic field lines. Plasma can be confined with the use of strong magnetic fields generated by current in coils, requiring a very specific geometry. Magnetic confinement is pursued in two types of devices: the Tokamak and the Stellerator.

These machines are made of a toroidal set of coils, which create a toroidal magnetic field topology. This geometry has numerous advantages for creating a confining magnetic field, however it also leads to certain problems, such as the radial decrease of the magnetic field. To counteract these issues, a poloidal field is needed to twist the field lines and create a helical geometry. These lines create nested toroidal magnetic surfaces, called toris, on which magnetic field lines are confined. In tokamak, the boundary of this confinement region is called the "separatrix".

This requirement of a poloidal field is at the basis of the difference between the two types of fusion devices. In Stellerator, the poloidal field is generated by the geometry of the coils which deforms the perfect toroidal geometry into a twisted torus, yet still topologically equivalent. In tokamak, the poloidal field is mainly generated by the current in the plasma, which is driven using an Ohmic system. For shaping the plasma equilibrium configuration, additionnal poloidal field (PF) coils are used. It modifies the geometry and are specifically useful to manage the lines that connect the confined region to the plasma facing components (PFCs). Contrary to the toris, for which the field lines close upon themselves, the outer magnetic field lines are open, meaning they are not bound to a closed region.

The PFCs are any structural element intentionally exposed to direct interaction with particles exiting the plasma confined region. This is at the core of one of the main challenge in creating long-term operationnal fusion devices, which is the power exhaust problem. As plasma reaches extreme temperatures, the wall are often subject to extreme load of heat flux. It can lead to serious damage of the material and decrease the lifetime of the device. In fusion devices, a "divertor" is used, which is a plasma facing system responsible for managing the heat flux. In the divertor region, situating outside the confinement region, the open magnetic field lines are specifically directed towards targets which are designed to support the heat load. The path to the wall is optimized and increased to maximize the dissipation of energy through radiation and recombination process. In EPFL, the "tokamak à configuration variable" (TCV), is known for its capability of generating a broad variety of plasma shape. It enables the study of alternative divertor configurations, such as the "Snowflake", the "Super-X"[1] or the "Jellyfish"[2], in which particles are deviated in numerous divertor target.

Studying particle behavior in this region can be challenging, as direct measurements are impossible inside the plasma. Alternative and sophisticated diagnostics are therefore required to measure densities and temperatures in the divertor region. On TCV, several measurement systems are implemented and actively studied, such as the Thomson scattering diagnostic (TS) [3], infrared thermography (IR)[4], and the multispectral imaging diagnostic (MANTIS)[5]. These diagnostics rely on complex physical processes, but they can be very efficient and are used either for real-time measurements or for post-processing analysis. One particular diagnostic, which provides the main motivation for this study, is the Multispectral Advanced Narrowband Tokamak Imaging System (MANTIS).

This diagnostic relies on the acquisition of images of different spectral line emissivities originating from atomic de-excitation processes. It is capable of measuring up to ten different wavelengths corresponding to specific atomic line emissions. By applying tomographic inversion techniques, a two-dimensional poloidal reconstruction of the emissivity can be obtained. Using calibration procedures and photon emissivity coefficient (PEC) models, these measurements can provide information on local density and temperature profiles. Such images are particularly useful for validating theoretical plasma configurations and for divertor studies. In some configurations, such as the Jellyfish configuration, MANTIS measurements reveal unexpected spatial patterns, which motivate further investigation.

Figure 1: Raw MANTIS image for a Jellyfish configuration, patterns similar to tentacles can be observed

In some Jellyfish configurations, specific patterns resembling tentacles appear in the MANTIS diagnostic. As shown in fig.1, these structures consist of elongated and narrow regions of high emissivity. They originate from the high-field side (HFS) and extend towards the low-field side (LFS). These patterns are observed over timescales much longer than typical turbulent timescales, given the camera exposure time. This suggests that they are not associated with fast turbulent fluctuations, and makes an interpretation purely based on turbulence or transient MHD activity unlikely. One possible explanation for the formation of these structures is that they originate from magnetic field perturbations, such as error fields.

Error fields (EFs) are unintended non-axisymmetric perturbations of the magnetic field, caused by imperfections in the magnetic field coils or their power supplies. As a result, the actual magnetic configuration deviates from the ideal one. A common source of error fields is the misplacement of magnetic coils. In TCV, the displacement of the poloidal field (PF) coils has been investigated in previous studies by J-M.More (1998)[6] and F.Piras (2010)[7], who developed specific methods to estimate these displacements using magnetic field measurements. Certain types of coil displacements that remain axisymmetric do not significantly modify magnetic field line behaviour. However, non-axisymmetric displacements, such as a toroidal tilt of a PF coil surrounding the plasma chamber, can generate non-axisymmetric magnetic perturbations. These perturbations break the axisymmetry of the plasma configuration and can lead to the formation of complex magnetic field structures, such as magnetic tangles.

This curious name of a magnetic tangle, refers to an equally intriguing physical concept, which originates from Hamiltonian dynamical system theory. Because the magnetic field is conservative and divergence-free, magnetic field line trajectories can be described within a Hamiltonian formalism. In such systems, complex structures can emerge, and chaotic behaviour may arise.

Chaos theory describes deterministic dynamical systems in which very small differences in initial conditions can lead to drastically different trajectories. This strong sensitivity to initial conditions was first identified by H.Poincaré in his studies of celestial mechanics (1892)[8], and was later further developed by Lorenz (1963)[9]. Fusion devices are highly complex systems, and some of their properties are particularly sensitive to small perturbations. Magnetic field line trajectories, for example, are extremely sensitive to variations in the magnetic field.

In axisymmetric fusion devices, the presence of non-axisymmetric magnetic perturbations breaks the axisymmetry of the magnetic configuration and can lead to chaotic magnetic field line trajectories. This also degrades the ideal confinement properties of the magnetic configuration defined by the separatrix. As a consequence, some field lines can cross the confinement region in localized areas, known as turnstile lobes. This transport mechanism is referred to as turnstile transport and originates from the presence of magnetic tangles near the separatrix. Such non-axisymmetric perturbations can also be intentionally generated using resonant magnetic perturbation (RMP) coils, installed in devices such as DIII-D, where they are used to modify the edge magnetic topology [10].

In the case of error fields, it is important to control these perturbations, as they can lead to confinement degradation. The formation and effects of magnetic tangles in diverted tokamaks have already been investigated in several studies [11, 12]. However, in TCV, no rigorous experimental evidence of magnetic tangles has yet been reported.

The objective of this work is therefore to investigate their possible presence by combining magnetic field line simulations with experimental observations.

To compute magnetic tangles in TCV, magnetic field line simulations are performed using tools from dynamical system theory, such as Poincaré maps. This method consists in selecting a fixed poloidal cross-section and recording the successive intersections of magnetic field lines after each toroidal turn around the device. In this representation, different field line behaviours can be identified, including invariant magnetic surfaces and chaotic regions. Within this framework, magnetic tangles and turnstile lobes naturally appear, when a non-axisymmetric perturbation is added. By computing the displacements of the magnetic coils, the associated magnetic perturbations can be evaluated and superimposed onto the axisymmetric magnetic equilibrium. The objective is then to investigate whether the spatial structure of the resulting magnetic tangle is consistent with the patterns observed in the MANTIS diagnostic images.

This work is structured as follows. First, the MANTIS diagnostic is introduced, and images of selected TCV configurations are presented, in which magnetic tangles may be present.

Second, the field line Poincaré map is described, together with the concept of turnstile transport, which is responsible for the formation of such structures.

Then, magnetic coil displacements are computed using the same methodology as in previous TCV studies. The resulting magnetic perturbations are computed and added to the axisymmetric equilibrium configuration.

Numerical simulations of field line Poincaré maps are then performed using the resulting magnetic field for the selected discharges.

Subsequently, a comparison between the numerical simulations and the MANTIS diagnostic images is carried out in order to assess whether error-field-induced magnetic tangles can account for the observed Jellyfish tentacle structures.

Finally, parametric variations are performed to study the sensitivity of the turnstile lobe patterns and to investigate possible sources of discrepancies.

2 MANTIS and Jellyfish tentacles

Resolving the power exhaust problem in the fusion device is one of the significant challenges in constructing a durable reactor. Modeling the electron density n_e and temperature in the divertor region T_e has always been a challenging task. This is due to the lack of diagnostics in the region. The usual diagnostics are generally limited to Thomson scattering and Langmuir probe measurement, which are limited in their measurement coverage[13]. This problem motivates the development of further diagnostics, such as multi-spectral atomic line imaging. In TCV, it is called the Multispectral Advanced Narrowband Tokamak Imaging System (MANTIS)[5].

Diagnostic can be used as a real time input to control detachment, in the model-based control algorithm. Secondly, using other parameters and assumptions, it can be used to compute a high resolution, 2-D line emissivity profile using tomographic inversion. In case of studying the turnstile transport mechanism in TCV, it will be used as the main comparative tool to infer the 2-D density profile in the divertor region.

In the next paragraph, the basis for line emission diagnostic will be explained together with an exemple of raw images captured by MANTIS. In the second chapter, the tomographic inversion technique will be described. The specific shots for which tangles are suspected will be presented.

2.1 Line-radiation and Collisional-Radiative Models

When a de-excitation occurs in an excited atom, a photon is emitted. The frequency of the photon depends on the energy difference between the two bound states from which the electron transitioned [14]. $\Delta E_{n_i->n_j} = h\nu = E_{n_i} - E_{n_j}$. The emissivity $\epsilon_{n_i->n_j}$ ($\frac{\#\ photons}{s \cdot m^3}$) produced from this transition is given by the following equation: $\epsilon_{n_i->n_j}(t) = A_{n_i->n_j} n_i(t)$. With n_i the density of the ion species and $A_{n_i->n_j}$ the transition probability, which depends on other characteristic of the surrounding plasma. To infer quantities such as density n_e and temperature T_e using line radiation, collisional-radiative modeling (CRM) is used [13]. The CRM is a system of first order differential rate equation, describing evolution of atomic populations. Generally, excitation of the atom occurs with electron impact excitation (EX) and the electron recombination

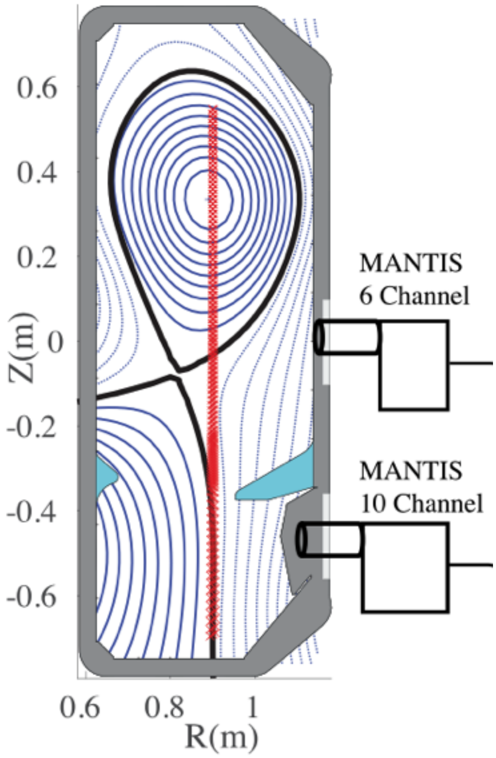


Figure 2: Poloidal section of the TCV with a schematic of the two MANTIS camera

process (RC). Other processes are also occurring; see ref.[13] for the Helium example. The transition probability then depends on the electron density n_e and the photo emission coefficients (PEC) of the excitation process. A common CRM emissivity model, which only accounts for RC and EX, is expressed by

$$\epsilon^{A}_{i->j} = n_A n_e PEC^{EX}_{i->j}(n_e, T_e) + n^{+}_{A} n_e PEC^{RC}_{i->j}(n_e, T_e) \ . \tag{1}$$

The data for the PEC indices are tabulated in the ADAS database. In order to infer T_e, n_e, a fit is typically applied using the measurements of the emissivity of various spectral lines of a specific atom.

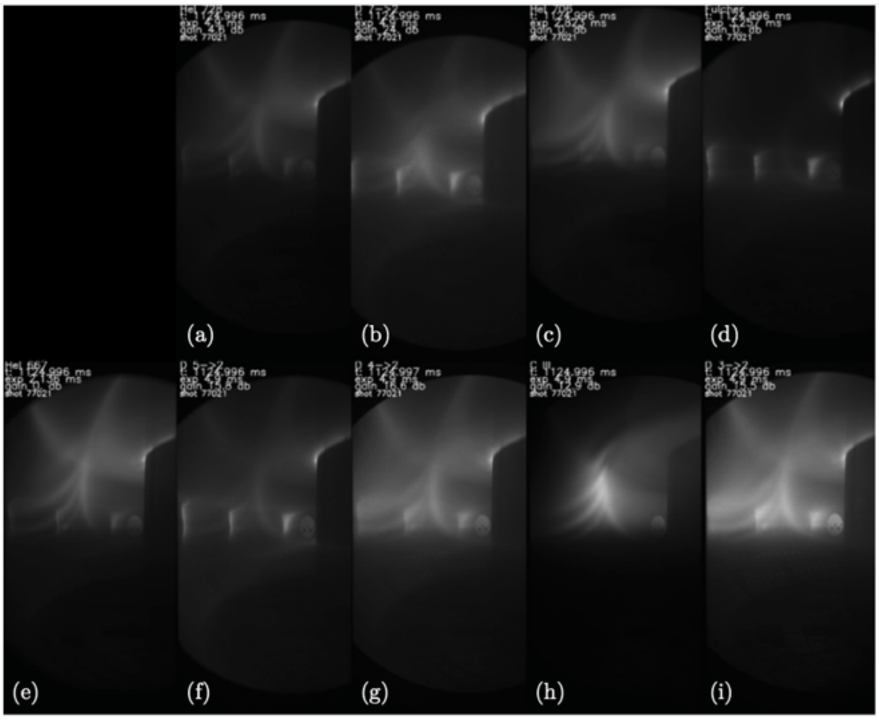


Figure 3: Raw image of different spectral line saw in the MANTIS cameras. The shot is the Jellyfish 77021 at 1.125 s. There are "tentacles", signs of potential magnetic tangles. Taken from S.Gorno's thesis (2024)[2]

In fig.3, raw MANTIS images of 9 different spectral lines are displayed for shot 77021; 3 Helium lines, 4 Hydrogen Balmer lines, the C-III line and the Fulcher. The Fulcher-α system consists of visible molecular emission bands of hydrogen and is commonly used for rotational and vibrational temperature diagnostics in plasmas. As can be seen, the emissivity profile depends on the considered spectral line. Each ionization process emits preferentially in a specific electron temperature and density regime, determined by the corresponding photon emissivity coefficient (PEC). Regions with high ionic densities n_A and n^{+}_{A} in the relevant excited states therefore contribute most strongly to the emission of a given line. In regions where the plasma temperature is too high, ions are predominantly found in higher ionization states, and the emission of lower-charge-state lines is reduced. In regions where the temperature is too low, the emission of higher-charge-state ions is reduced. In a tokamak with a divertor, spectral line emissivities strongly depend on the local neutral density, as neutrals provide the particle source for excitation, ionization, and recombination processes, particularly in the edge and divertor regions.

The MANTIS tool is sensitive to where neutrals and plasma interact; it is particularly efficient for providing the parameter profiles in the divertor region. In fig.3, stripes similar to "tentacles" are visible, which are typical of this specific configuration; the jellyfish. These patterns are the suspected turnstile lobes, which arise from a magnetic tangle. They are visible in most of the lines, but appear more distinctly in the C-III line in fig.3h. The camera captures the raw signal at each pixel, which comes from a convoluted portion of the plasma. To compute the emissivity profile in a poloidal section, a technique called the tomographic inversion reconstruction is used. In the next chapter, the diagnostic will be described with a description of the inversion technique.

2.2 Inversion and tomography

The MANTIS system contains two cameras located at two different heights, as shown in fig.2. Each camera has a polychromator with a different number of channels: 10 channels for the bottom camera and 6 for the second. In a camera's optical cavity, it uses narrow passband filter of approximately 1.5[nm][5]. A specific pixel of the camera p_i^m measures the integrated emissivity of a monochromatic light source $L^m(\nu_0)$, and its expression is given by

$$b_i = p_i^m \frac{\gamma_i}{f(\nu_0)} \int_L \epsilon(\mathbf{r}) dl \ . \tag{2}$$

With $f(\nu_0)$ the transmission of the bandpass filter and γ_i a calibration parameter. To perform the tomographic inversion, the geometric transfer matrix $K_{ij} = ds_{ij}$ is used. This matrix relates the i^{th} emissivity ray in the inversion grid to what the j^{th} pixel sees [15, 13]. The emissivities are then computed by assuming toroidal symmetries and solves the following expression

$$\mathbf{b} = \underline{\underline{K}} \cdot \epsilon \tag{3}$$

using the Simultaneous Algebraic Reconstruction Technique (SART). The MANTIS frame rate usually operates from a few 100 Hz to 1000 Hz depending on the brightness of the line. The time exposition is on the order of a few [ms] to hundreds of [ms] depending on the brightness. The tomographic reconstruction for the C-III line of shot 77021 is shown in fig.4a. Two other jellyfish's shot are also displayed in fig.4b and 4c. The poloidal profile of the emissivity is reconstructed using tomographic inversion.

As the process reconstruct the signal using computational technique, it has uncertainties that can lead to image artifacts production. They often come from the assumption of toroidal symmetry, which is sometimes - during this work, for example - not verified. There is also a horizontal V-shaped artifact that is often visible, visible in fig.4c, which does not come from localized radiation but from the localization of the camera[5]. Certain lines can also be polluted by surface reflection or other atomic line which have a similar frequency. The tomographic reconstruction has to be analyzed visually, because not all aspects of the image come from radiation activities.

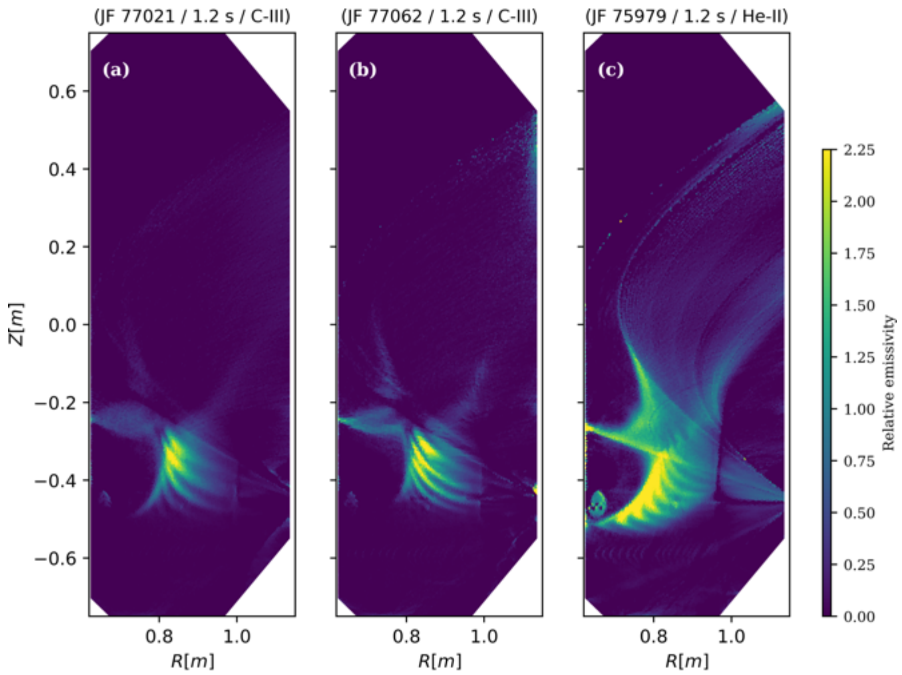


Figure 4: Tomographic reconstruction of three Jellyfish shots with visible tentacles in the separatrix region.

The MANTIS tomographic reconstructions are available in the matlab code in the LAC server. Although it was not always computed for the shots of interest, data were still measured during these shots. Some tomographic inversions were specifically computed for this work by Ricardo Ian Morgan.

In this work, the C-III (465 [nm]) and He-II lines (468 [nm]) will mostly be used. Generally, hydrogen balmer lines are often used, but the tomographic reconstructions from these lines are often polluted by surface reflection in the shots of interest. The lines from which the images are the cleanest and are used as density maps of the divertor region to compare with turnstile patterns. Absolute calibration is not required, as density patterns are sufficient to identify high density region. Each line could potentially give us a correct 2-D poloidal density map. However, depending on the atomic line used, the emissivity profile can vary.

The C-III line usually shows high emissivity along the divertor leg and near the separatrix. This is caused by the fact that its emission usually originates from the high temperature region where C^{2+} still appears [5]. It is showed in fig.4a and fig.4b.

He-II has a broader emissivity; it has a strong n_e dependence and is less sensitive to electron temperature in the typical range of the divertor region. In a B-L.Linehan's study (2023)[13], results showed a good convergence between density and temperature computed from Thomson scattering data and tomographic reconstruction. A He-II tomographic reconstruction is shown in fig.4c.

He-II and C-III can show a front emission along the divertor leg. This indicates a rapid decrease in temperature and/ or density. C-III, specifically, can quickly transition from an emitting state to a non-emitting state at temperature and densities observed in this region. It is often used as a low-temperature indicator. This can be observed in fig.4a, where the lower divertor leg is located. Properties regarding spectral line emissivities will be further discussed, namely in sec.7, when results will be analyzed.

The motivation for this work, originated from images captured by this diagnostic. The specific stripes are typical signs of magnetic tangle and the reasons are described in the next chapter.

2.3 Signs of magnetic tangle

As shown in fig.3 and fig.4, regions of high emissivity are captured in specific structures, which will be referred to as "tentacles" or "legs" in this work. They are typically observed in a specific type of plasma magnetic equilibrium configuration, namely the jellyfish configuration. As shown in the tomographic reconstruction in fig.4, these structures depart from the separatrix region, extend along the divertor leg, and point towards the HFS. These patterns are visible and nearly stationary from approximately 1.0 s to 1.4 s in the MANTIS camera. Since the exposure time is typically 5 ms for MANTIS and the characteristic turbulent timescale is of the order of 10 μs [16], the possibility that the tentacles are due to turbulent MHD activity can be ruled out.

Patterns of this type are a typical signature of the presence of a magnetic tangle. Such a structure originates from non-axisymmetric perturbations, which can be caused, for example, by displacements of the poloidal field coils. When axisymmetry is broken, the ideal confinement properties of the plasma equilibrium degrade. The separatrix then allows transport of particles from the confined region to the scrape-off layer through the turnstile transport mechanism. This process creates a set of small lobes containing particles originating from the confined plasma, arranged in a complex pattern near the divertor region, referred to as a magnetic tangle. The objective of this work is to demonstrate that the tentacle structures observed by the MANTIS diagnostic align with theoretical predictions of turnstile lobes. To compute the turnstile lobe patterns, field-line tracing simulations are required. These are performed using a Poincaré map, which will be described in the next chapter.

3 Poincaré map

Magnetic field line trajectories can be described within the framework Hamiltonian dynamics. In toroidal confinement devices, their evolution can be formulated as a Hamiltonian system with $1\frac{1}{2}$ degrees of freedom, where the phase space is typically associated with suitable geometric coordinates [17, 12]. This formulation allows the application of well-established tools from Hamiltonian dynamics, including variational principles and action-based methods [18, 19].

Direct numerical integration of three-dimensional magnetic field line trajectories is, however, computationally expensive, particularly when long connection lengths or large ensembles of field lines are required. For this reason, reduced descriptions based on discrete mappings are frequently employed. A central tool in this context is the Poincaré map of magnetic field lines.

Because magnetic fields are divergence-free, the associated field-line flow preserves phase-space volume. When restricted to a two-dimensional surface of section, the induced dynamics can be represented by a two-dimensional symplectic, area-preserving map. Replacing the continuous three-dimensional description with such a discrete map therefore reduces the dimensionality of the system while retaining its essential invariant dynamical properties. It provides a powerful framework for the study of chaotic systems, offering a clearer insight than the full 3-D description. With this tool, we will be able to describe complex field-line topology and transport in a TCV plasma equilibrium configuration. It will then enable the study of the turnstile transport, which is the suspect for creating the divertor tentacles, as described in sec.2.

The poincaré map captures the intersection of the field-lines at a certain ϕ plane location. As tokamaks are axisymmetric, in contrast to stellarators, the field lines are commonly captured at every single toroidal turn. Rather than relying on analytical mapping functions, which are often impractical or computationally demanding for magnetic fields generated by external coils, most studies employ field line integration techniques. These methods numerically integrate the magnetic field-line equations along the toroidal angle and record the intersections with the chosen Poincaré section at each toroidal circuit.

In this chapter, the basis of the magnetic field lines Poincaré map will be described: the magnetic line trajectories, the fixed-points and their manifolds.

3.1 Field line map

Following field lines can be described by an integral curve $\mathbf{c}(t) : \mathbb{R}_+ \to \mathbb{R}^3$, written in cylindrical coordinates as $\mathbf{c}(t) = (R(t), \phi(t), Z(t))$. A field line is a curve whose tangent vector is the magnetic field $\frac{d\mathbf{c}}{dt} = \mathbf{B}(\mathbf{c}(t))$. In the coordinate basis (R, ϕ, Z), this reads $\dot{R} = B^R$, $\dot{\phi} = B^\phi$ and $\dot{Z} = B^Z$, where B^R, B^ϕ, B^Z denotes the contravariant components of $\mathbf{B}$. It is often convenient to make a parameter change by replacing t with the toroidal angle ϕ. This is allowed by the fact that $B^\phi \neq 0$ along the curve, making the ϕ coordinate monotonic. With $dt/d\phi = 1/B^\phi$ and the chain rule, the curve is now

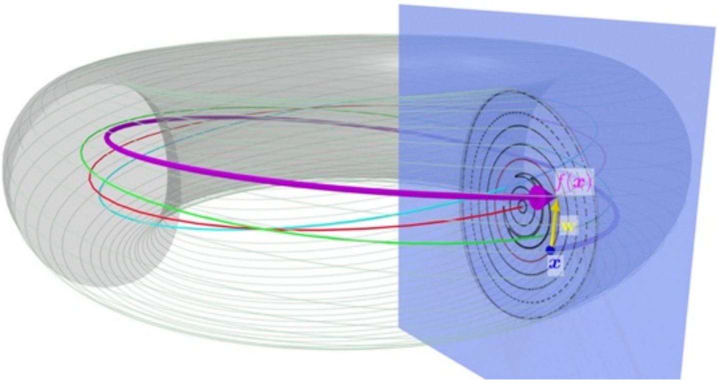


Figure 5: Representation of a field line Poincaré map, typically in a toroidal fusion geometry. A point $\mathbf{x}$ is mapped to $\mathbf{f}(\mathbf{x})$, following the magnetic field lines for one turn. The red curve represent a fixed-point $\mathbf{x}^*$, which is mapped to itself after 1 iteration, $\mathbf{f}(\mathbf{x}^*) = \mathbf{x}^*$

given by $\mathbf{c}(\phi) = \mathbf{c}(t)\frac{dt}{d\phi} = \frac{\mathbf{c}}{B_\phi}$ and the full explicit reparametrization given by

$$\dot{\mathbf{c}}(\phi) = \left(\frac{B^R(R(\phi), \phi, Z(\phi))}{B^\phi(R(\phi), \phi, Z(\phi))}, 1, \frac{B^Z(R(\phi), \phi, Z(\phi))}{B^\phi(R(\phi), \phi, Z(\phi))} \right) . \tag{4}$$

By using the coordinate ϕ as the curve parameter, the field-line dynamics are reduced to the (R, Z) plane. The poincaré map is a discrete mapping between successive intersections of a field line with a given Poincaré section U, defined by $f : U \to U$: $(R_{k+1}, Z_{k+1}) = \mathbf{f}(R_k, Z_k)$. In cylindrical coordinates, $U \in (\mathbb{R}_+, \mathbb{R})$. For a tokamak, which is axisymmetric, the 1-mapping maps points in the section to their next intersection with the same section after following the field lines for $N = 1$ toroidal turns $\phi_{k+1} = \phi_k + T,\ T = 2\pi N,\ N = 1$. In non-axisymmetric configurations such as stellarators, the return period generally depends on the number of field periods n_{fp}, and the Poincaré map is typically defined after $N = 1/n_{\text{fp}}$ toroidal turns. The Poincaré map, with initial condition (R_k, Z_k) at the toroidal section ϕ_k, is explicitly given by

$$(R_k, Z_k) \to \mathbf{f}(R_k, Z_k) = \int_{\phi_k}^{\phi_k+T} \left(\frac{B^R(s)}{B^\phi(s)}, \frac{B^Z(s)}{B^\phi(s)} \right) ds + (R_k, Z_k) . \tag{5}$$

A schematic of a typical magnetic field lines Poincaré map in a fusion devices is shown in fig.5. The Poincaré section is represented in blue and four 3-D trajectories in red, green, magenta and cyan. An iteration corresponds to a full toroidal turn of the field lines. The map can also be iterated m times according to $\mathbf{f}^m(R_k, Z_k) = \overbrace{\mathbf{f} \circ \mathbf{f} \circ ... \circ \mathbf{f}}^{m}(R_k, Z_k) = (R_{k+m}, Z_{k+m})$. The representation of the Poincaré map provides a powerful tool for studying the qualitative behavior of the field lines. It can reveal the underlying phase-space structure, such as invariant closed surfaces, magnetic island chains and chaotic regions.

Strange and complex patterns emerge when the system is in a chaotic state. In fusion devices, this typically occurs when non-axisymmetry is introduced into axisymmetric equilibrium by adding magnetic field perturbations. An example of a typical poincaré plot with its complex and beautiful patterns is displayed in fig.6. This is generated using an analytical large ratio tokamak with perturbed fields. The crosses and dots represent fixed points, which are central to the study of the dynamical features of the trajectories of field lines. The four concentric structures visible in fig.6 represent bounded regions, which surrounds a specific type of fixed point called "O-point".

A fixed-point $\mathbf{x}^*$ is defined as a point which is mapped onto itself after m iterations. $\mathbf{f}^m(\mathbf{x}^*) = \mathbf{x}^*$. A schematic of a fixed-point trajectory is represented by the red trajectory in fig.5. Their properties are discussed in the next chapter. The fixed-points of Poincaré are actually the 2-D equivalent of periodic orbits. This is a typical example in which the use of a 2-D map can simplify the representation of complex 3-D objects, such as closed orbits. Strogatz mentioned them in his book as:

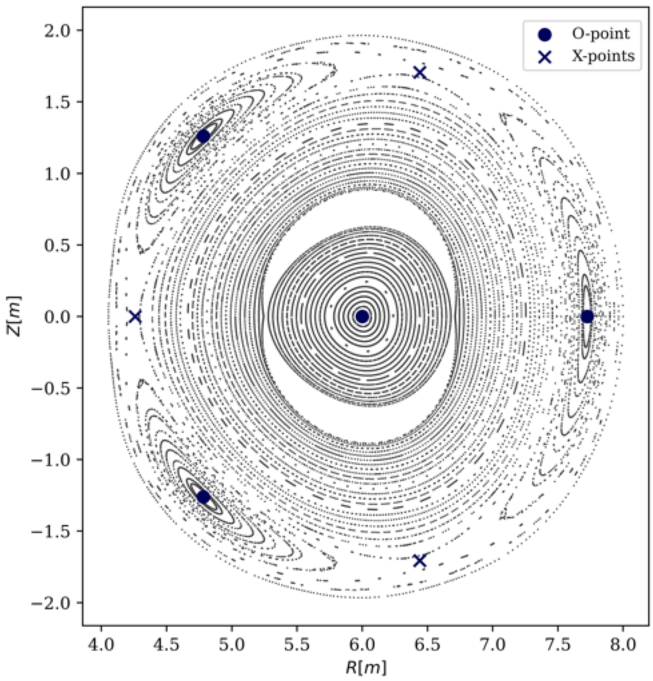


Figure 6: Poincaré plot of a high aspect ratio analytical tokamak configuration. A non axisymmetric perturbation is added, which creates typical complex patterns of a chaotic dynamical system.

> "Thus the Poincaré map converts problems about closed orbits (which are difficult) into problems about fixed points of a mapping (which are easier in principle, though not always in practice)" [20]

3.2 Fixed points

To find fixed points, numerous algorithm are used such as Newton type methods. In the package `pyoculus` used here, the `scipy.root` method is mainly used [21]. The dynamical behavior of field lines around fixed points exhibits characteristic structures, which can be determined by linearizing the field line poincaré map around the fixed point. A first order Taylor expansion of the field-line Poincaré map is performed in the neighborhood of a fixed point and is given by

$$\mathbf{f}^n(\mathbf{x}^* + \epsilon) \simeq \mathbf{f}^n(\mathbf{x}^*) + \mathbf{M}^n \cdot \epsilon \ . \tag{6}$$

The matrix $\mathbf{M}^n$ is the Jacobian of the Poincaré map given by eq.(5), $M^{ij} = \frac{\partial f^i}{\partial x^j}$ with $\mathbf{x} = (R, Z)$. The subscript n denotes the order of the map. It corresponds to the number of toroidal turns required for a fixed point $\mathbf{x}^*$ to return to itself. ϵ is an infinitesimal displacement.

The divergence free condition $\nabla \cdot \mathbf{B} = 0$, which guarantees flux conservation and implies that the Poincaré maop is area-preserving, leads to the Jacobian matrix satisfies $\det(\mathbf{M}^n) = 1$. The matrix $\mathbf{M}^n \in SL(2, R)$ is then an element of the special linear group, which consists of 2-by-2 real matrices with determinant 1. This group is a connected non-compact simple real Lie group. The eigenvalues of $\mathbf{M^n} \in SL(2, R)$ matrices are solutions of the characteristic equation $\lambda^2 - tr(\mathbf{M}^n) \cdot \lambda + 1 = 0$, with the following solution,

$$\lambda = \frac{1}{2}\left(tr(\mathbf{M}^n) \pm \sqrt{tr(\mathbf{M}^n)^2 - 4}\right) \ . \tag{7}$$

Elements of $SL(2, \mathbb{R})$ can be classified into three distinct types according to the value of the trace, each corresponding to a different local phase-space structure around the fixed point:

- $|tr(\mathbf{M}^n)| < 2$, M is called elliptic. The eigenvalues form a complex conjugate pair with unit modulus, and the linearized dynamics correspond to a rotation. The associated fixed point is referred to as an *O-point* and is surrounded by invariant closed curves, which correspond, in this context, to invariant magnetic surfaces. This is shown in fig.7a and in fig.6.
- $|tr(\mathbf{M}^n)| = 2$, M is parabolic. This category has less interest in our case.
- $|tr(\mathbf{M}^n)| > 2$, M is hyperbolic. Eigenvalues are both real. The fixed point is called an X-point and is associated with intersecting stable and unstable manifolds, which locally form hyperbolic structures in phase space. The eigenvector for which $|\lambda_s| < 1$ is called the stable direction $\mathbf{e_s}$ and $|\lambda_u| > 1$ the unstable one $\mathbf{e_u}$. The most common kind of hyperbolic fixed-point, which is of interest in this study, is the regular hyperbolic point, shown in fig7b. It is defined by the property $tr(\mathbf{M}^n) > 2$. Eigenvalues are both positive. Trajectories approach or depart from the fixed point monotonically along the stable (in blue) and unstable directions (in red).

The two types of fixed points that are of primary interest for studying Poincaré maps in fusion devices are O-points and X-points.

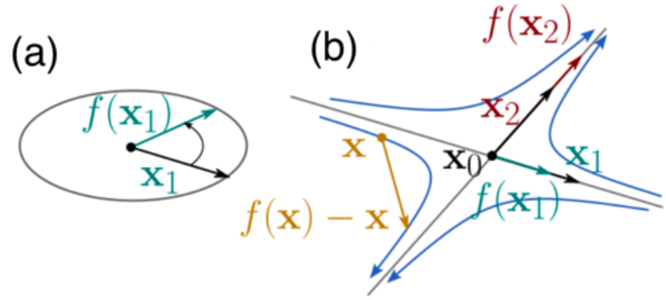


Figure 7: Schematic of two kinds of fixed-points. (a) elliptic point, "O-point" (b) hyperbolic point, "X-point". In red, point and its mapping situated on the unstable direction. In blue, on the unstable direction. In yellow, on a hyperbola. Taken from Smiet et al. (2020)[22]

The confined region of the plasma corresponds to a bounded phase-space formed by invariant tori surrounding an O-point. In plasma jargon, it is also referred to as the magnetic axis. The last closed flux surface (LCFS) is the last invariant toroidal surface, which has the largest area.

For X-points, there are two directions corresponding to two eigenvectors $\mathbf{e_s}$, $\mathbf{e_u}$ of the matrix $\mathbf{M}$. In the vicinity of the fixed-point and along the stable directions $\mathbf{e_s}$, $\mathbf{x} = \mathbf{x}^* + \mathbf{e_s}\epsilon$, points will converge to the fixed-point by iterating forwardly $\mathbf{f}$. Along the unstable direction $\mathbf{e_u}$, $\mathbf{x} = \mathbf{x}^* + \mathbf{e_u}\epsilon$, points converge to the fixed-point by iterating backwardly $\mathbf{f^{-1}}$. This property of convergence is not restricted to points displaced infinitesimally around an X-point. It is generalized in objects called manifolds. The stable manifold W^s is defined as the ensemble of points that converge to x^* for an infinite number of forward iterations $n \to \infty$, formally expressed by

$$W^s(x^*) = \{x | f^n(x) \to x^* \ as \ n \to \infty\} \tag{8}$$

and the unstable manifold W^u for an infinite number of backward iterations $n \to -\infty$, given by

$$W^u(x^*) = \{x | f^n(x) \to x^* \ as \ n \to -\infty\}. \tag{9}$$

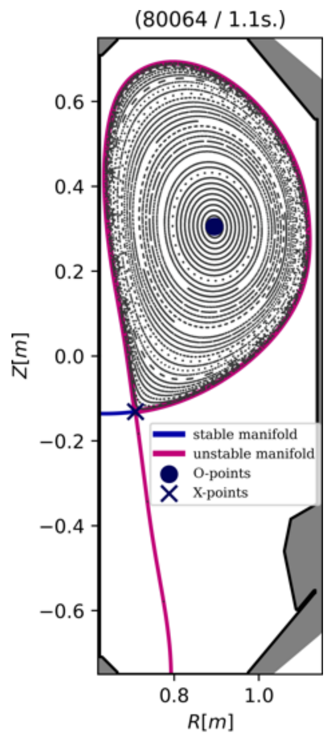


Figure 8: Poincaré plot for a non-perturbed TCV single null-equilibrium.

These objects are particularly interesting for studying invariant surfaces. In case of axisymmetric plasma configuration, they form a perfect boundary surface arround the invariant toris surrounding the magnetic axis. They are superposed and no transport is allowed between the inside and the outside.

To visualize this phenomenon, a non perturbed single null TCV configuration is displayed in fig8. The O-point is at the center of a constant invariant region, formed by nested magnetic surface and surrounded by a superposed stable and unstable manifolds derived from the hyperbolic X-point. The X-point have 2 stable and 2 unstable manifolds, depending on the direction of the eigenvector $\pm\mathbf{e_s}$ and $\pm\mathbf{e_u}$. The manifolds directed upward form the separatrix and the other two manifolds are connected to the walls. The LCFS that surrounds the confined region is the largest outermost magnetic flux surface composed entirely of closed magnetic field lines. In the case of axisymmetric configurations, the LCFS converges to the separatrix.

The surface inside the curve formed by the separatrix is bounded, constant and there is no flux from the confined region to the outside region. There is also no chaos. When perturbations are added, or in a more general case when chaos is present, the situation becomes a little more complicated. This will be discussed in the next chapter. The separatrix is still defining the confined region, but the manifolds are no longer superposed. They allow transport from the inside to the outside in a process called turnstile transport.

4 Chaotic regime

In the real world, perfect axisymmetry does not exist. Small non-axisymmetries in the coils break the ideal axisymmetric configuration; the resulting perturbation fields are commonly referred to as error fields (EFs). These perturbations give rise to new physical phenomena, such as the formation of magnetic islands and the onset of chaotic magnetic field line behavior.

In the presence of non-axisymmetric perturbations, the invariant magnetic flux surfaces that were previously bounded by the separatrix are broken, and chaotic transport processes emerge. Among these processes, turnstile transport plays a key role. This mechanism connects magnetic field lines from the open field-line region to the confined plasma region through narrow structures known as turnstile lobes. Such transport processes can lead to enhanced particle and heat losses, making the control of magnetic chaos essential in order to preserve plasma confinement and limit heat fluxes to the plasma-facing components.

These phenomena have been extensively studied in poloidally diverted tokamaks [12, 17]. These perturbations can also be created by Resonant Magnetic Perturbation coils (RMP) to manually create perturbations [11].

In this chapter, the processes leading to turnstile transport will be described. Firstly, the effect of introducing a non-axisymmetric perturbation will be discussed through the analysis of an analytical example. Secondly, homoclinic orbits will be introduced, as they are necessary for the description of turnstile transport, which will be the subject of the last part of this chapter.

4.1 Effect of a non-axisymmetric magnetic perturbation

In this chapter, a perturbed analytical configuration will be simulated to discuss the emergence of chaotic field lines and its effect on an equilibrium configuration.

Perturbations in analytical cases can be simulated by adding a periodic perturbation to the ϕ component of the vector potential $A^{\phi} = A^{\phi} + \epsilon\delta A$. ϵ is the amplitude factor, and the perturbation is expressed in terms of a Fourier series $\delta A = \delta f(\rho)\cos(n\phi + \alpha_n)\cos(m\theta + \alpha_m)$. n is defined as the torodial mode and m the poloidal mode. A perturbation is characterized by its (m,n) Fourier modes. The term $f(\rho)$ is the radial distribution function, which depends on the toroidal radius ρ and the distance to the magnetic axis. It can represent any distribution function, such as the maxwell-boltzmann distribution or a normal distribution.

In fig.9a, a Poincaré section is presented for a single null analytical field, along with the addition of a separatrix field. In this case, there is no perturbation field, and it is axisymmetric. Similar to the TCV example in fig.8, the Manifolds delineate an invariant and bounded region composed of closed field lines that surround the magnetic axis. The nested concentric tori can be described as surfaces of constant poloidal field flux function ψ. In this case $\psi = R^2 A^{\phi}$, but in real tokamak case it is more common to have $\psi = RA^{\phi}$. The description in terms of ψ are useful, as the poloidal field component can be derived solely from this function, requiring a choice of gauge $A^R = 0$ and $A^Z = 0$. This choice is permissible considering the axisymmetry of the device. Here, magnetic field lines lie on constant tori, and no field lines are connected to the outside region. All of these field lines are closed, meaning they remain confined to a given flux surface.[1]

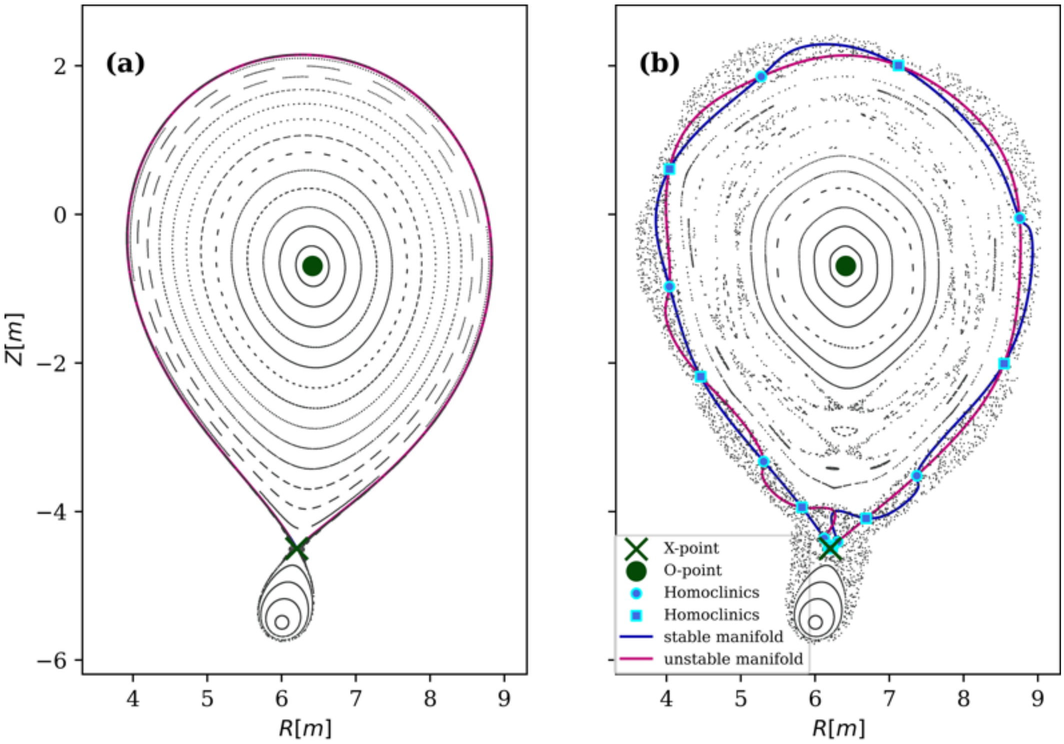


Figure 9: Analytical field high aspect ratio tokamak. In left, no perturbation. In right, a MB perturbation with n=1 and m=6. Chaos has been introduced.

In fig9b, the same configuration is displayed, but this time with the addition of a (6,1) Maxwell Boltzmann perturbation. The perfect axisymmetry and the perfectly nested toroidal magnetic surfaces are now broken in some regions. There is now a chaotic region, for which a small displacement in phase-space leads to a completely different trajectory. These regions are typically located between the magnetic island chains and near a broken separatrix. A magnetic island inside region, which typically emerge from non-axisymmetric perturbations, is also referred to as a resonance zone. More information are provided about these resonance zone in A.2.Here, the description is limited to turnstile transport, which is the only process required to compute the turnstils lobes.

Now that a periodic perturbation has been introduced, the stable and unstable manifolds surrounding the O-point confined zone are no longer superimposed, as demonstrated in fig.9b. This results in the formation of small lobes, referred to as turnstile lobes, between segments of stable and unstable manifolds. Inside these lobes are field

[1]The strict definition of a field line's closure is that field lines return to themselves after a certain number of toroidal turns; however, this definition is commonly accepted

lines connected to the external region. In blue, there are points of intersection of the manifolds, which are referred to as clinic orbits. They are important for the turnstile transport as they delimit the lobe region. They are defined in the next chapter.

4.2 Homo-heteroclinics orbit

Since the manifolds are no-longer superimposed, they intersect only at certain points, represented by the blue point in fig.9b. They are called homoclinic or heteroclinic orbits. These are important for studying chaotic transport as they delimit the turnstile lobes. A clinic point m is described as a point that intersects an unstable manifold W^u and a stable one W^s, expressed by

$$m \in W^u(h_1) \cap W^s(h_2) \ . \tag{10}$$

If the two manifolds W^u/W^s converge (backwards/ forwards) to different fixed points $h_1 \neq h_2$, $p \in W^s(h_2)$, $lim_{n\to\infty} f^n(p) = h_2 = h$, $p \in W^u(h_1)$, $lim_{n\to-\infty} f^n(p) = h_1$, the ensemble of points m that intersect the two manifolds are called heteroclinic orbits.

If the two manifolds converge to the same point, $h_1 = h_2 = h$, they are referred to as homoclinic orbits. For the broken separatrix of a confined region composed of a separatrix and an O-point, there exist homoclinic orbits.

For now we will keep the homoclinic term as this is the type of clinics present around a tokamak confined region.

Different types of homoclinic orbits exist. The ones that are of great interest for studying chaotic transport are primary homoclinics. A homoclinic m is primary if the only intersection between the segments $\partial S = m - h,\ S \in W^s(h)$ and $\partial U = m - h,\ U \in W^u(h)$ occurs at their endpoints [19].The primary homoclinics are important as they delimit the turnstile lobes. They are found by a root-finding algorithm and are described in C.Smiet et al. work (2025)[23]. Once a primary homoclinic is found, applying forward and backward iteration leads to the trajectory of m_i, which also represents homoclinic points. However, there is at least one other primary homoclinic trajectory to find, as turnstile lobes are delimited by at least two different homoclinic trajectories.

Now that the homoclinics are defined, the turnstile transport process can be described, which is the content of the next chapter.

4.3 Turnstile lobe transport

The turnstile transport is a special type of transport that occurs in Hamiltonian dynamical systems described by measure-preserving maps. It is described in great detail in the works of MacKay et al. (1984)[18] and Meiss (2015)[19]. The magnetic field line Poincaré map is an area-preserving map, meaning that the area of a closed region C is conserved under iteration Area(C) = Area($f^n(C)$). In the case of a magnetic field line Poincaré map, this property is directly related to the conservation of magnetic flux ϕ through a surface that is transverse to the magnetic field, given by

$$\phi(R) = \int\int_R \mathbf{B} \cdot d\mathbf{n} = \int\int_{f^n(R)} \mathbf{B} \cdot d\mathbf{n} \ ,\ \forall n \in Z \ . \tag{11}$$

The flux crossing a closed curve bounding a region R of phase-space, $C = \partial R$, is defined as the area A that is mapped outside after one iteration $f(R) = R'$. The area is defined as $A = R'/R$. As magnetic field lines mapping is $1\frac{1}{2}$ Hamiltonian with its phase space being the geometric coordinates, the flux represents the incoming and outgoing guiding center flow within the delimited curve. For an invariant closed curve topologically equivalent to a circle, such as the non-perturbed manifolds around an O-point in fig.9a, the region delimited by the separatrix is constant when it is iterated backwards and forwards. This implies that there is no flux crossing the region. It forms a complete barrier to motion. The whole region is perfectly confined as every field lines are closed inside this region.

For a perturbed field, such as fig.9b, the pair of manifolds forming the separatrix around the O-point are now "broken", the bounded region is no longer invariant. This implies that there is a flux crossing the region. They form a partial barrier, meaning that there is flux that can cross the region, requiring numerous iterations. As the flux is conserved by the divergence free or, equivalently, the area-preserving property, the incoming flux and the outgoing flux must be equal. In the turnstile lobe are magnetic field lines that are connected to the outside region. These lobes act like a rotating door (hence the name), as for each iteration, the lobe of the exit set E is pushed outside, and the lobes of the incoming set I are pushed inside.

In fig.10, the turnstile transport process for a pair of broken separatrix is shown. The bounded region R can be defined as the area enclosed by the curve formed by two segments of stable and unstable manifolds $\partial R = U + S$. $\partial U = h - m,\ S \in W^u(h)$. $\partial S = m - h,\ S \in W^s(h)$. The point m is an arbitrary primary homoclinic point, and the point h is the separatrix fixed-point, represented by the red cross.

The exit set is defined as the region E that escapes the closed region after one iteration, $E = \{x \in R | f(x) \neq R\}$ [19]. It is represented by the E lobe in fig.10b. The backwardly iterated version of E, such as $f^{-1}(E)$ and $f^{-2}(E)$ are located inside the delimited confined zone. The forward iterated version of E, are in the contrary outside of the delimiting zone, as the blue stable segment $\partial S = m - h$ delimits the region. The more it is iterated the more it is pushed further outside, as it can be seen with $f^4(E)$. As every lobe have the same area, they can all be considered as being the exiting lobe.

The incoming set I is the set of points in phase-space that just entered the closed region $I = \{x \in R | f^{-1}(x) \neq R\}$. It is represented by the I lobe in fig.10a. The lobe formed by the backward iteration $f^{-n}(I)$ are located outside of the arbitrary delimiting curve, for which the unstable segment was defined as $\partial U = h - m$. Its images are mapped inside, and its pre-images are located outside. These lobes intersect field lines situated in the outside region, allowing transport from the outside to the inside of the bounded region. The separatrix is now a partial barrier. The turnstile is the region where the door mechanism occurs, defined as $T = E \cup f^{-1}(I)$.

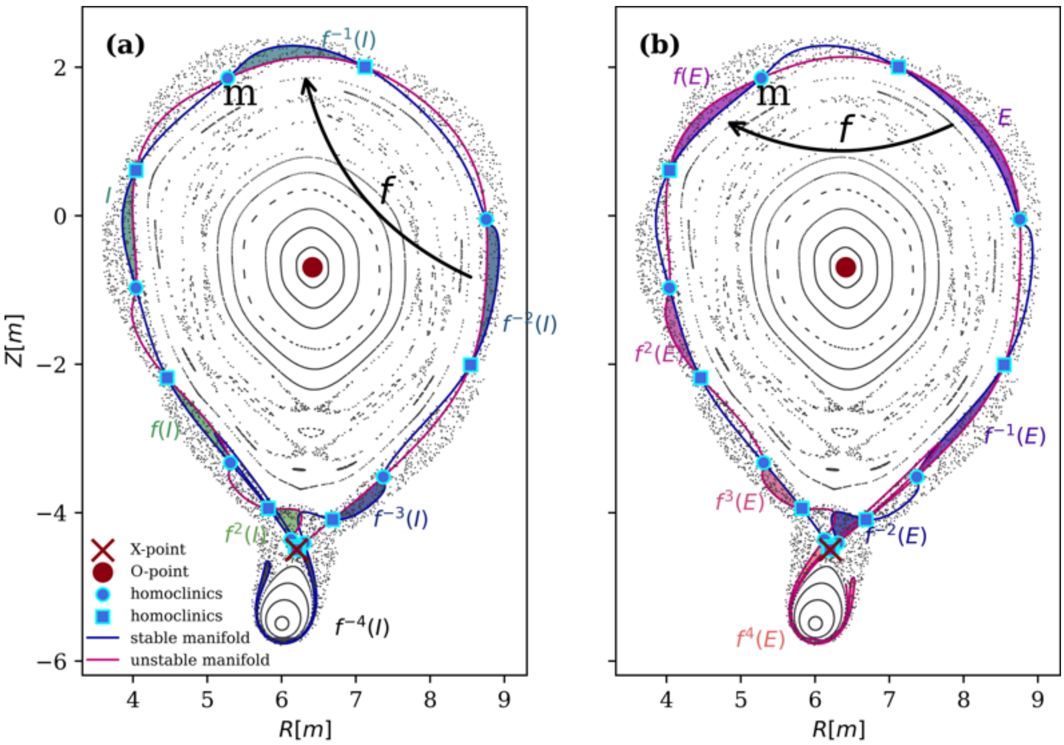


Figure 10: Turnstile transport process in a perturbed analytical example. In (a) the entry process and in (b) the exit process.

The number of primary homoclinic points is infinite, as they can be iterated arbitrarily far both forward and backward under the map. These points converge exponentially toward the hyperbolic fixed point, approaching it along the unstable manifold under backward iterations and along the stable manifold under forward iterations. Consequently, the distance between two successive homoclinic points, such as m and h, becomes exponentially small. As the magnetic flux enclosed within a turnstile lobe is conserved, according to eq.(11), the lobes are increasingly stretched as they approach the fixed point. As shown in fig.10, $f^{-4}(I)$ and $f^{4}(E)$ exhibit highly elongated shapes. The more the turnstile lobes are iterated backward and forward, the more they are stretched and folded. This process generates a complex structure in the vicinity of the X-point, referred to as a tangle. In the case of a split separatrix associated with a single X-point, this structure is referred to as a homoclinic magnetic tangle, since both the stable and unstable manifolds originate from the same hyperbolic fixed point.

Turnstile transport is associated with chaotic transport, as an infinitesimal displacement in a chaotic region, such as the region close to the X-point when non-axisymmetric perturbations are present, can lead to a completely different trajectory. In particular, a field line may enter a different turnstile lobe belonging to the same homoclinic or heteroclinic tangle.

An example of an error-field magnetic tangle in TCV (which will be explained in the next chapters) in a single-null configuration is shown in fig.11. Near the X-point separatrix, a homoclinic tangle is formed as a consequence of the separatrix splitting induced by the error field. The closer the turnstile lobes are to the X-point, the more they are stretched and folded around the separatrix. H.Poincaré already observed such structures in his work on celestial mechanics (1892)[8], where he described the complex folding associated with homoclinic intersections. These patterns were later studied in detail by R.W.Easton (1986)[24], who referred to them as a trellis because of their complex and interwoven structure. The folding of turnstile lobes is a typical example of chaotic dynamics involving stretching and folding mechanisms. The Smale-Birkhoff theorem states that the presence of a homoclinic tangle implies a dynamical behaviour that is topologically equivalent to a horseshoe map [25], in which stretching and folding naturally occur. This provides a useful framework for interpreting the complex structures observed in magnetic tangles.

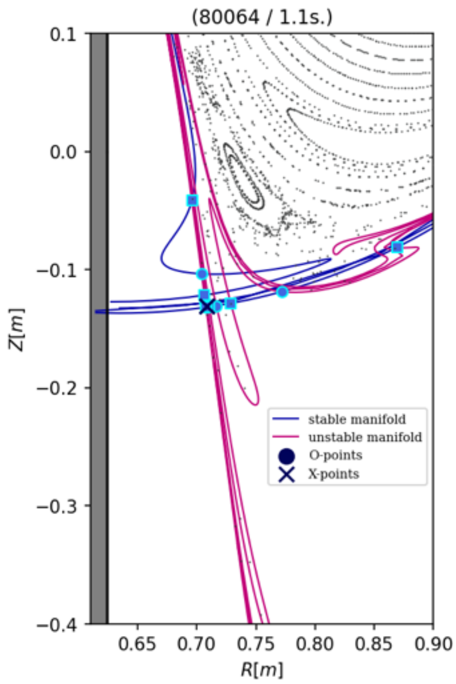


Figure 11: Magnetic homoclinic tangle in a perturbed TCV single null-equilibrium.

To control the flux of particles that cross a bounded region of phase-space, computing the turnstile lobe or a resonance zone magnetic flux can be useful. However, direct computation can be computationally costly, especially for non-analytical cases. To make these calculations more practical, action principles derived from the Hamiltonian flow measure-preserving map properties are used. The turnstile lobe calculation using Meiss principle was already implemented in `pyoculus` in L.Rais's thesis [26]. Here the implementation of the resonance zone area calculation using the same action principle is done and is shown in A.2.

In this chapter, the turnstile transport was described. Inside the turnstile, are magnetic field lines which are connected to the outside. It allows magnetic flux to enter through the entry set and to exit through the exiting set. The turnstile lobe properties and patterns seem to correlate with the high emissivity stripe patterns visible in the MANTIS diagnostic. They are situated near the X-point and the fact that high emissivity is recorded, suggest high temperature and ions/ electrons densities which are typically observed in the confined plasma region. The non-axisymmetric perturbation in TCV, which would cause a magnetic tangle, could be induced by the non-axisymmetric placement of PF coils. The first step to compute the error-field is to compute their displacements. The method which rely on certain measurement diagnostic will be the subject of the next chapter.

5 Error-field in TCV

One of the suspects in the creation of the specific patterns visible in certain Jellyfish shots is the induced error-field. These fields are generated by various uncertainties in the different coils and diagnostics. The hypothesis of this study is that these patterns represent the turnstile lobe formed by error-field perturbation manifolds. To observe these structures, a non-axisymmetric perturbation with a resonance of n=1 must be present. The primary candidate that meets these specific criteria is the displacement of the PF coils relative to their intended location. Since PF coils are expected to be perfectly axisymmetric, any displacement of the center or a tilt will induce a non-axisymmetric perturbation. To measure these displacements, magnetic measurements are employed. The calculations for each type of coil and probe displacement have been previously conducted by J-M.Moret (1998)[6] and F.Piras (2010)[7]. A similar procedure is applied, but with a few simplifications. Only the displacement that causes an n=1 error field will be computed, because the n=0 error field does not create a non-axisymmetry perturbation. They only affect the plasma shape, position, and degradation of the null point at the breakdown[7]. Also, a few simplifications are made in order to adapt the theory to the available matlab TCV tools.

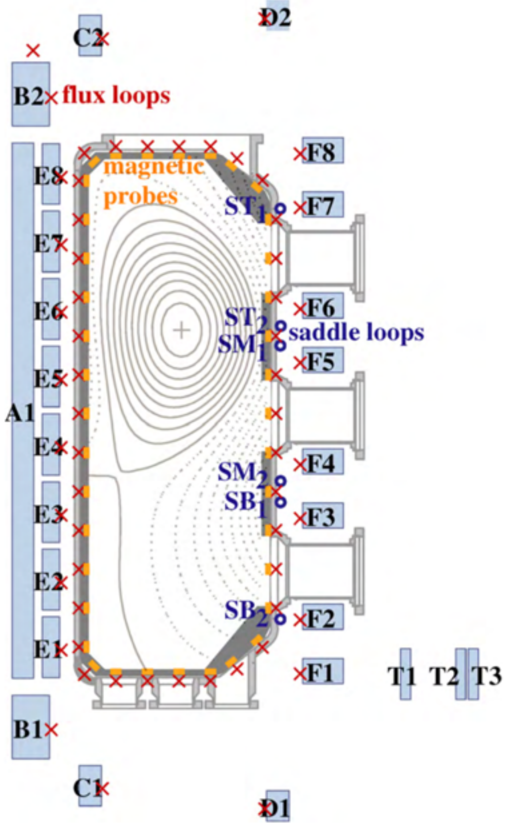


Figure 12: Section of the TCV with a schematic of the poloidal field coils and the measurement systems.

5.1 PF Coil system

The TCV poloidal field coil system contains 3 different sectors: E,F and OH as shown in fig.12. The E coils are the inner field coils, the F coils are the outer field coils, and the OH coils are the ohmic heating induction coils. The OH1 system contains coil A. The OH2 system contains coils B1, B2, C1, C2, D1 and D2 which are connected in series. The 8 E coils and the 8 F coils serve as the plasma shaping coils.The toroidal field coils are not taken into account in this study, as their high number and the toroidal component are less likely to generate a n=1 resonant magnetic field perturbation. However, if the vacuum confinement zone is displaced relative to the structure containing the toroidal coils, it could create a resonance perturbation of n=1.

5.2 Magnetic measurements

Magnetic measurement in TCV consists of three different measurement systems, as shown in fig.12. The poloidal flux loops, the magnetic probes and the saddle loops are presented.

- The poloidal flux loops are represented by the red crosses. They are placed in front of the PF coils, and they measure the poloidal flux. They will not be used in this study, as they are not sensitive to the types of displacements that are relevant for computing the non-axisymmetry-induced error-field.
- The magnetic probe system is composed of 2 sectors of 38 probes inside the vacuum vessel, represented by orange points. The first sector b_3 is located in sector 3 $\phi_{b_3} = 67.5$.The second b_{11}, is located in sector 11 $\phi_{b_{11}} = 247.5$. The units are in Tesla $[T]$.
- The saddle flux loops (ψ_s) , which are rectangular circuit loops covering 45deg of toroidal angle. The system contains 24 SL's. It is divided into 4 sections of 6 SL's, 2 columns of 3 stacked loops, denoted BOTTOM, MIDDLE, TOP. As they measure magnetic flux, their units are in Weber $[Wb]$.

5.3 Calculation of displacement

To compute the center shift and tilts, a diagnostic that is spatially distributed and sensitive to small variations in the poloidal field is needed. The optimal measurement system exhibiting the best characteristics is the saddle loop system. The magnetic probes was also used at the beginning, but they proved to induce too much uncertainties for their inclusion. This is discussed in A.1. For now, only saddle loops are considered.

The calculation is based on a F.Piras study (2010)[7]. In order to infer the coil displacement, a first order expansion is performed on the magnetic flux measured in the saddle loops ψ[wb]. The saddle loops measure the voltage induced by the time-derivative of the magnetic field. They can also be integrated to obtain the flux itself. The theoretical magnetic flux induced by the current in the PF coils is given by the following expression: $\psi_{\mathbf{s}}^{\mathbf{th}} = \underline{\underline{\mathbf{M_{sc}}}}\mathbf{I_c}$. With $\underline{\underline{\mathbf{M_{sc}}}}^{48\times18}$ the matrices of Green's function[27]: the mutual inductance between the respective instrument and the current (I) in the PF coil system is given by

$$M_{ij} = -\frac{1}{4\pi\epsilon_0 c^2}\oint_i \oint_j \frac{d\overrightarrow{s_i}\, d\overrightarrow{s_j}}{r_{ij}} \ . \tag{12}$$

Here, Einstein summation is used to simplify, with the index c representing the coil index and s denoting the saddle loop index. The coil current value $\mathbf{I_c} = \underline{\underline{\mathbf{T_{cw}}}}\mathbf{I_w}$ is, in fact, related to $\mathbf{I_w}$, the current variable of each loop current, and $\underline{\underline{\mathbf{T_{cw}}}}$, the transfer matrix between the individual loops and the coil. To simplify the notation, only the terms of the current value $\mathbf{I_c}$ are retained.

The measurements in the saddle loop contain uncertainties, which can be estimated by

$$\Delta\psi_{\mathbf{s}} = \psi_s - \underline{\underline{\mathbf{M_{sc}}}} I_c \ . \tag{13}$$

They will be our input values for inferring the displacement values of the coils. To obtain the expression of the error field caused by the displacement of the coils, a first order expansion is applied to $\psi_{\mathbf{s}}^{\mathbf{th}}$. The whole computation, including other terms, can be found in ref. [7]. Here, only terms that cause a perturbation of (n=1) are retained. These terms include the displacement $\mathbf{\Delta X}^{1\times18}$ and $\mathbf{\Delta Y}^{1\times18}$ of the centers of the coils, as well as the $\mathbf{\Delta}\alpha_{\mathbf{x}}{}^{1\times18}$, $\mathbf{\Delta}\alpha_{\mathbf{y}}{}^{1\times18}$ tilts of the coils, and the linearized expression for $\Delta\psi_s$ is given by:

$$\Delta\psi_{\mathbf{s}} = (\frac{\partial \underline{\underline{\mathbf{M_{sc}}}}}{\partial X_c}\mathbf{\Delta X_c} + \frac{\partial \underline{\underline{\mathbf{M_{sc}}}}}{\partial Y_c}\mathbf{\Delta Y_c} + \frac{\partial \underline{\underline{\mathbf{M_{sc}}}}}{\partial \alpha_{cx}}\mathbf{\Delta}\alpha_{\mathbf{cx}} + \frac{\partial \underline{\underline{\mathbf{M_{sc}}}}}{\partial \alpha_{cy}}\mathbf{\Delta}\alpha_{\mathbf{cy}})\mathbf{I_c} \ . \tag{14}$$

In this expression, assumptions such as the small displacement regime, localized angle measurement, and dominance of the coil effect in a localized angle prevent the need for a complicated functional derivative of eq.(12). To convert to cylindrical coordinates, as this is the system that the Green's function in matlab utilizes, the chain rule is applied. For displacement X, $\frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial X_c} = \frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial r_c}\cos\theta$. Utilizing the dominance of localized coil portion approximations, the trigonometric average $< \cos(\theta_s) >= \frac{1}{\theta_{s2}-\theta_{s1}}\int_{\theta_{s1}}^{\theta_{s2}}\cos(\theta')d\theta'$ to account for the loop distribution, and the substitution $\partial_{r_s} = -\partial_{r_c}$ as there is no coil center derivative in the existing toolbox, gives the following expression $\Delta M \simeq -\frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial r_s} < \cos\theta_s > \mathbf{\Delta X_c}$. The last assumption holds since a displacement in X for the coil is equivalent to a displacement of the center of the saddle, for the small displacement approximation. θ_{s2} and θ_{s1} are the maximal and minimal angles of a certain s loop. The coordinates (r_s, z_s) are the location of the different saddle loops centers.

For tilts, the small angle approximation makes the change of coordinates dependent only on ∂_z. For the α_{oy} tilt, the z location of the coil loop at angle θ is given by $Z = \cos\theta\sin\Delta\alpha_{oy}$, utilizing the positive tilt convention along the ox axis. Then, using the dominance of a localized portion of the coil, the chain rule, the small angle approximation, and the substitution $\partial_{z_c} = -\partial_{z_s}$, the following expression is obtained. $\frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial \alpha_{cy}}\mathbf{\Delta}\alpha_{\mathbf{cy}} \simeq -\frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial z_s} < \cos(\theta_s) > \cos\mathbf{\Delta}\alpha_{\mathbf{cy}} R_c \mathbf{\Delta}\alpha_{\mathbf{cy}} \simeq -\frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial z_s} < \cos(\theta_s) > \mathbf{\Delta_t Z_c^x}$. The tilt $\Delta\alpha_{oy}$ can be expressed as the Z displacement it makes along the X axis $\mathbf{\Delta_t Z_c^x}$. The entire equation for the magnetic diagnostic components induced by the non-axisymmetry error-field can be expressed,

$$\begin{gathered}\Delta\psi_{\mathbf{s,i}} = -\frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial r_s} < \cos\theta_s > \mathbf{I_{cc,i}}\mathbf{\Delta X_c} - \frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial r_s} < \sin\theta_s > \mathbf{I_{cc,i}}\mathbf{\Delta Y_c} \\ -\frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial z_s} < \cos\theta_s > \mathbf{I_{cc,i}}\mathbf{\Delta_t Z_c^x} - \frac{\partial \underline{\underline{\mathbf{M}_{sc}}}}{\partial z_s} < \sin\theta_s > \mathbf{I_{cc,i}}\mathbf{\Delta_t Z_c^y} \equiv \underline{\underline{\mathbf{A}}}^{\psi}\mathbf{x} \ .\end{gathered} \tag{15}$$

The index i has been added, indicating the index of a measurement.

The system is then $\underline{\underline{\mathbf{A}}}\mathbf{x} = \mathbf{b}$ with $\underline{\underline{\mathbf{A}}} = \underline{\underline{\mathbf{A}^{\psi}}}$ and $\mathbf{b} = \Delta\psi_{\mathbf{s}}$. The displacement vector to be found is $\mathbf{x} = (\mathbf{\Delta X}, \mathbf{\Delta Y}, \mathbf{\Delta_t Z^x}, \mathbf{\Delta_t Z^y})$. The double c subscript in $\mathbf{I_{cc,i}}$ indicates a diagonal matrix (18x18) with the current in each coil. It is used to ensure that the matrix expression $\underline{\underline{\mathbf{A}}}\mathbf{x} = \mathbf{b}$ holds.

To approximate these values, a weighted least squares fitting with L2 regularization is applied. Regularization and weighting are added to prevent excessively large displacement values.

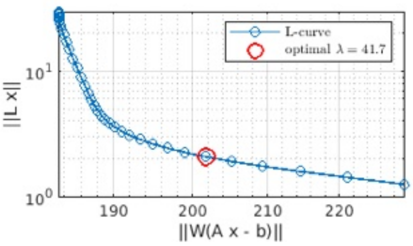


Figure 13: L-curve for the fitting of the coil displacement calculation. Optimized hyperparameter $\lambda_{optimal} = 41.7$

The weights are determined by the inverse of the inaccuracy of the saddle loop measurement $W = \delta m^{-1}$, which is $\delta\psi = 1[mW]$ [28]. The cost function is defined as follows:

$$L(\mathbf{x}) = ||W(\underline{\underline{\mathbf{A}}}\mathbf{x} - \mathbf{b})||^2 + \lambda^2||\mathbf{x}||^2 \ . \tag{16}$$

The size of the system depends on the number of measurements taken n: $\mathbf{I}^{18\times n}/\ \psi^{48\times n}$.

With λ being a hyperparameter, its value must be properly chosen. To this end, the L-curve method is applied; see fig.13. The value of λ providing the best trade-off between data fidelity and the size of the regularized solution corresponds to the point of maximum curvature of the L-curve, i.e. the plot of the solution norm versus the residual norm [29]. It gives the optimized order of magnitude for this hyperparameter. The solution is obtained by minimizing the cost function given in eq.(16), and the solution can be computed accordingly [30],

$$\mathbf{x} = (\mathbf{X^T X} + \lambda\mathbf{I})^{-1}\mathbf{X^T y} \ . \tag{17}$$

With $\mathbf{X} = \mathbf{WA}$ and $\mathbf{y} = \mathbf{Wb}$, the uncertainty associated with the fitted parameters is estimated from the ridge regression covariance matrix. The uncertainty used in the following is taken as the square root of the variance, which is the square root of the diagonal elements of the ridge covariance. The covariance matrix of the ridge regression is given by[30]:

$$Cov(\mathbf{x}) = \sigma^2(\mathbf{X^T X} + \lambda\mathbf{I})^{-1}\mathbf{X^T X}(\mathbf{X^T X} + \lambda\mathbf{I})^{-1} \tag{18}$$

The error bars used in the following correspond to $\sqrt{\mathrm{diag}(Cov(\mathbf{x}))}$.

5.4 Backoff and stray shots

To obtain a fitted solution for the displacement estimations, specific data samples are used. Two specific types of shot can be used for this computation. The first kind are strayshots. This type of shot is made for coil performance checks, where each coil is turned on at different timings during a single shot. The other kind is a set of shots, which is called backoff shots. During a backoff shot, one single coil is turned on during the entire shot. Since there are 18 coils, 18 separate shots are used. The current profiles of the PF coil currents are shown in fig.28. As shown, the coil currents in the strayshot has less time to stabilize, which is expected to provide less reliable data for the displacement fitting. In addition, the current amplitudes in strayshots are smaller, which may also lead to a noisier fit.

For comparison, 2 different strayshots and 2 different backshot sets are used. The strayshots are conducted regularly, typically every morning of shots. However, data for them are not always available. Backoff shots are performed less often; they are performed after extensive work on the tokamak. Two sets were conducted during the years 2020-2025. Theoretically, coils are supposed to be located similarly during all of these shots. However, there are other factors that could affect the movement and the results of the displacement fitting. For each peak in stray. and shot in backoffs, 10 different current measurements $(\mathbf{I}, \psi)$ are sampled, which results in a total of $n = 180$ samples. The number can also be increased to reduce the error bar, but the results do not differ significantly.

- Strayshots 72377, date: 14/12/2021
- Strayshots 80460, date: 26/03/2024
- Backoff shot set 78005-78076, date: 09/05/2023
- Backoff shot set 82993-83075, date: 09/03/2024

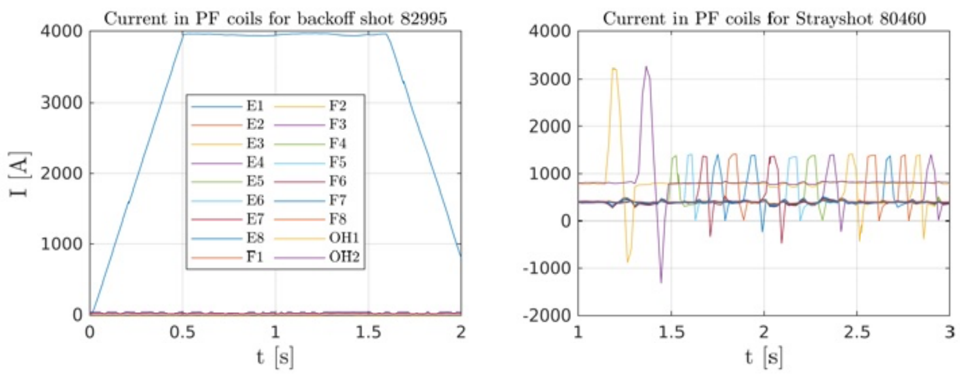


Figure 14: PF Coil currents in a backoff shot and a strayshot. On the left, current in each coil for E1 backoff shot 82995. Only coil E1 is turned on at 4000 [A] during 1 [s]. On the right, PF coils current in a strayshot, each coils are turned on at different time, during peaks of 10-20 [ms].

5.5 Comparison of coil displacement results

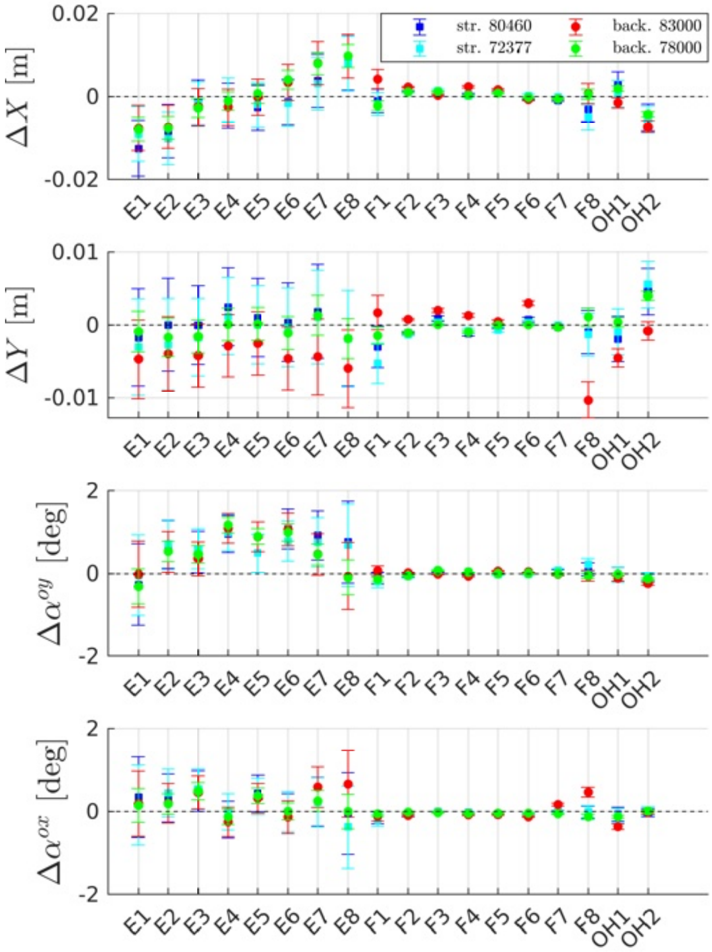


Figure 15: PF coil displacements and tilts computed with the fitting of eq.(15). Four different datasets from 4 different shots/ shotsets are used.

For ΔX, as shown in fig.15, the four results are situated within the same order of magnitude and exhibit a similar trend. They all lie inside $\{-0.01; 0.01\}[m]$. The larger displacements are in the E coils, with a maximum displacement of 1 [cm]. The error bars are larger for the E coils. This could arise from the fact that the E coils are located farther away from the saddle loops than the F coils.

For ΔY, the results are within $[-0.005; 0.005][m]$, with a maximum displacement of 0.5 [cm]. The matches between results are correct for the 3/4 dataset. For backoff 83'000 (red circle), the results are below for E coils, above for F coils, and there is a significant discrepancy for F8. It is difficult to determine whether this general discrepancy is due to a genuine displacement in ΔY or to fitting and other factors that may have altered the measurements.

For $\Delta\alpha^{oy}$, the results are on the order of the degree. The matches exhibit generally good agreement across the different datasets. The largest tilts are also for the E coils.

For $\Delta\alpha^{ox}$, the results tend to be smaller than $\Delta\alpha^{oy}$. There is also a small mismatch between the backoff 83000 (BO83) and the other three results.

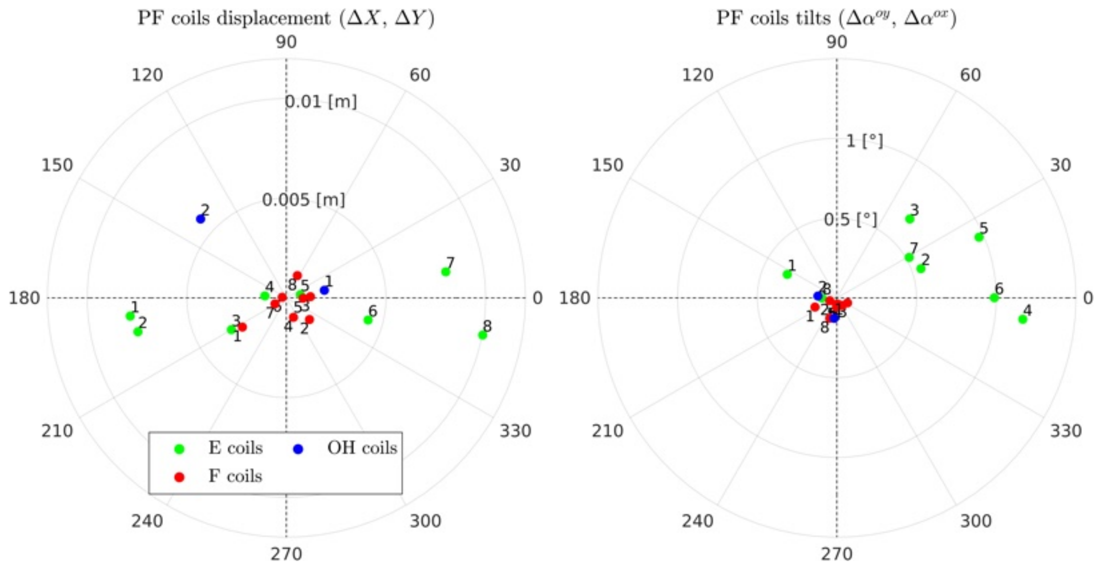


Figure 16: Displacements in and tilts computed with BO78 in a 2-D plot. In left, PF coils displacement in X and Y. In right, OX and OY tilts

The displacements that seem to be most suited for computing the error-field in the discharges used in this study are those computed from BO78. The shots for which the error field will be computed are those around shot number 76000. Thus, BO78 is the closest to them.

It also has the smallest error bar. Since it is computed from backoff shots, the measurements are more time-stabilized than those for strayshots. However, the impact on the inaccuracy is not substantial for strayshots, as the error bars lie within the same order of magnitude. The coils did not seem to move during shots 72377 and 80460. However, there may have been some coil movements that occurred between 80460 and 83000, as discrepancies exist for BO83.

Displacement and tilts (from BO78) are shown in fig.16 as a 2-D plot to enhance the understanding of the general displacements.

The central columns (E coils) seem to have a general tilt in $\Delta\alpha^{oy}$. In the tilt plot, the small angle approximation $\beta \simeq \arctan(\frac{\Delta\alpha^{oy}}{\Delta\alpha^{ox}})$ is used. The location of the point to the origin is the axis for which the ΔZ caused by the tilt is maximum and positive. β is the angle of this axis and OX.

There is also a general center displacement that increases with the height of the E coil located along the OX axis. This could be caused by a tilt of the central column. This would cause the center of the different coils to exhibit a center displacement dependent on their height location.

The same conclusion was drawn in Piras's work [7] regarding the central column being shifted. The values for ΔX in this work were found to be between $[-0.01, 0.01][m]$ and his work $[-0.001, 0.005][m]$. He also found a larger value for ΔY. The tilts have the same profile regarding the coils, but results are smaller. It could be caused by the different fitting process, or a genuinely different coil position. This discrepancy could stem from differing weights used in the fitting process. This matter was not further investigated in this study.

In fig.17, the values for the error-field-induced magnetic field distribution for a toroidal section for the R and Z components of the magnetic field are plotted. There is a large contribution to the perturbations coming from the E4 coil in the HFS at $Z \simeq -0.1[m]$. The perturbations are on the order of milli-Teslas. As the perturbation is in the toroidal field mode n=1, it is 2π toroidally periodic. Now that a set of displacements has been computed, the final step is to compute the magnetic field perturbation they generate in TCV configurations. This topic will be addressed in the subsequent section. Firstly, the equilibrium of a tokamak configuration is discussed alongside the code LIUQE, which provides the axisymmetric equilibrium magnetic field configuration in TCV. The perturbations from displacements that cause error-fields are then presented along with a couple of numerical steps of the coupling LIQUE-`pyoculus` and the first results.

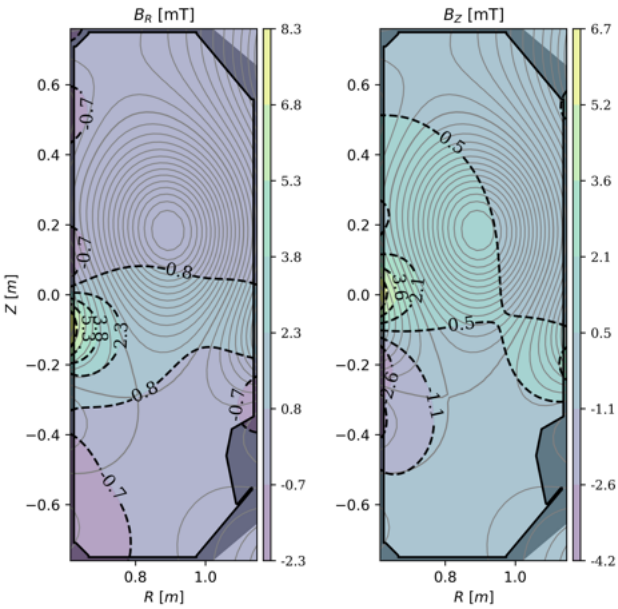


Figure 17: Error-field induced magnetic perturbation distribution for shot 75979, at 1.2s. Left B_R and right B_Z

6 Turnstile in TCV

We have seen how non-axisymmetry can cause magnetic stochastic field lines and turnstile transport. We have also shown that the poloidal field coils are not perfectly axisymmetric and computed their displacements. Now we can put this together and compute the perturbed Poincaré plot for TCV plasma configurations.
The first step is to describe the axisymmetric plasma equilibrium configuration in tokamaks. It is computed by solving a well-known equation called the Grad-Shafranov equation. The code LIUQE, which solves this equation and provides axisymmetric datas in TCV, will be discussed. Then, the routine for computing Poincaré maps using the LIUQE-`pyoculus` coupling will be described. Finally, the magnetic field perturbations caused by the non-axisymmetry of coils and the way they are combined with the axisymmetric equilibrium configurations will be described. The first result of perturbed Poincaré map in TCV will be discussed.

6.1 Grad-Shafranov and LIUQE

In an axisymmetric configuration, the plasma components are obtained by solving the Grad-Shafranov equation, which governs ideal MHD equilibrium in axisymmetric magnetic confinement devices. This equation is derived from the 3 equations that govern ideal MHD equilibrium[14][31]. They are the divergence free equation (Gauss's law) shown in eq.(19), the Ampère's law in (20) and the momentum equation in eq.(21):

$$\nabla \cdot \mathbf{B} = 0 \ , \tag{19}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} \ , \tag{20}$$

$$\mathbf{j} \times \mathbf{B} = \nabla p \ . \tag{21}$$

Using cylindrical coordinates (R, ϕ, Z), along with the axisymmetry of a torus $\partial_\phi = 0$ and Gauss's law, the system can be simplified by deriving the magnetic field from a potential $\nabla \times \mathbf{A} = \mathbf{B}$. Using $RA_\phi = \psi$, the following expression is obtained:

$$\mathbf{B} = -\frac{1}{R}\frac{\partial \psi}{\partial Z}\mathbf{e_r} + \frac{1}{R}\frac{\partial \psi}{\partial R}\mathbf{e_z} + F(\psi)\mathbf{e}_\phi \ . \tag{22}$$

The system now depends on the poloidal flux function $\psi(R, Z)[wb]$ and the function $F(\psi) = RB_\phi$. The function ψ can also be expressed as the total poloidal flux function per radian $\psi_{pr} = \frac{\psi}{2\pi}[wb/rad]$. Now, substituting $\mathbf{B}$ components into the Ampère's law and then the resulting current density $\mathbf{j}$ into the force balance, the Grad-Shafranov equation is obtained and given by the following expression:

$$\mu_0 R^2 \frac{\partial p}{\partial \psi} + F\frac{\partial F}{\partial \psi} + \Delta^* \psi = 0 \ . \tag{23}$$

For more details, see Wesson's book [31]. With $\Delta^* = \frac{\partial^2}{\partial R^2} - \frac{1}{R}\frac{\partial}{\partial R} + \frac{\partial^2}{\partial Z^2}$ the elliptic operator. This equation needs to be solved to reconstruct the magnetic equilibrium inside an axisymmetric fusion device. It provides equilibrium solutions in which the outward plasma pressure gradient is balanced by magnetic forces, namely magnetic tension and magnetic pressure.

The surfaces of constant ψ define the surface on which magnetic field line trajectories lie. The magnetic field lines are tangent to these surfaces, and due to the MHD assumptions (eq.(19)-(21)), they are also constant current and pressure surfaces. The ψ function is the total poloidal flux, poloidal field components are obtained through differentiation and a specific choice of gauge,

$$(B^R, B^Z) = \left(-\frac{1}{2\pi R}\frac{\partial\psi}{\partial Z}, \frac{1}{2\pi R}\frac{\partial\psi}{\partial R}\right) . \tag{24}$$

To construct Poincaré plots of tokamak magnetic configurations, the two key quantities required are the poloidal magnetic flux ψ and the toroidal field function $F(\psi) = RB_\phi$. The total poloidal magnetic flux ψ is commonly decomposed into a plasma contribution ψ_p and a coil contribution ψ_C,

$$\psi(R, Z) = \psi_p(R, Z) + \psi_C(R, Z) , \tag{25}$$

where the plasma contribution ψ_p arises from the magnetic field generated by the plasma toroidal current, while the coil contribution ψ_C originates from the currents flowing in the external poloidal field coils [32, 33]. The coil contribution is entirely computed from the coil currents using the Biot-Savart law,

$$\psi_C(R, Z) = \sum_{i=1}^{N_c} I_i\, G(R_i, Z_i; R, Z) , \tag{26}$$

where (R, Z) are the coordinates of the evaluation point and (R_i, Z_i) are the radial and vertical positions of the i-th coil. Here, G denotes the Green's function relating the current in coil i to the poloidal flux at the location (R, Z). As discussed in Sec. 5, this Green's function corresponds to an element of the matrix $\underline{\underline{M}}_{xc} \equiv G(R_i, Z_i; R, Z)$.

In practice, ψ_C is treated as a prescribed external contribution, either through boundary conditions or as a known source term, while ψ_p is determined self-consistently through the Grad-Shafranov equilibrium equation.

In the fixed-boundary model, where the last closed flux surface (LCFS) remains fixed, the coil flux function is incorporated into the Grad-Shafranov equation as a boundary condition, with eq.(25) specified on $\partial\Omega$ and Ω representing the inner plasma domain. In a more general formulation, the free-boundary condition is used. In this case, the plasma boundary is not maintained as fixed, and ψ_C contributes throughout the entire domain formed by the vacuum and the plasma. The LCFS is then defined as an isoflux surface of the poloidal flux.

In TCV, one of the algorithms that solves the Grad-Shafranov equation under free-boundary conditions is called LIUQE[34]. It provides the data ψ, F, p which are solutions to the equilibrium supposing perfect axisymmetry. The LIUQE code will not be explained in too much detail; it utilizes also other magnetic diagnostic measurements to compute the equilibrium configuration, which can be used in real time. Here, it will be used to furnish the axisymmetric equilibrium data to feed the `pyoculus` code and compute Poincaré plots.

6.2 LIUQE equilibrium reconstruction to `pyoculus`

In this chapter, we will describe how we generate datas using LIUQE and feed it to `pyoculus`. LIUQE can be run on any specified TCV shots. It solves the GS equation and furnishes the ψ and B^{ϕ} variables for the axisymmetric equilibrium on a discrete grid of a toroidal section. $(R^{1\times 28}[m], Z^{1\times 65}[m], B_{\phi}^{28\times 65}[T], \psi^{28\times 65}[wb])$. The R-Z grid is defined in cylindrical coordinates as $(R, Z) \in [0.614, 1.147] \times [-0.76, 0.76]$. In fig.18, the green line represents the constant ψ surfaces computed by LIUQE for shot 75979.

To compute the Poincaré plot of the field-line tracing derived from a real TCV shot, the `pyoculus` package in python is used. It was commonly used for stellarator fields and analytical fields; thus, it is adapted to work on a TCV configuration. It must be working now on a discrete grid of ψ functions, provided by our GS reconstruction code. The data grids are extracted from the TCV matlab LAC server and transferred to python. The R-Z values of the magnetic field are given by eq.(24). To continuously compute the magnetic field values, a quintic spline interpolator is used on the 2-D (R, Z) grid [35]. The interpolation function can also provide derivative values. After implementing the magnetic field components, the Jacobian, and the integrated vector potential, the instance of `pyoculus` can work.

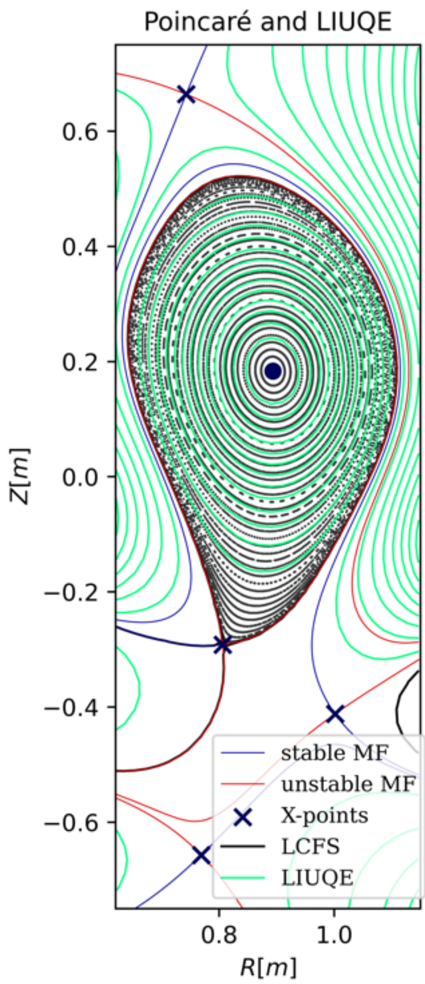


Figure 18: Poincaré plot of the LIUQE axi-symmetric equilibirum of shot 75979 at 1.2 s.

In fig.18, a Poincaré/ LIUQE plot is presented. As a field line mapping follows magnetic field lines, which lie on a constant ψ surface, both should coincide, and this is indeed confirmed numerically. The nested surfaces formed by the Poincaré trajectories coincide with the constant ψ surface generated by LIUQE. The manifolds correspond to the separatrix and the last closed flux surface (LCFS).

Here, no perturbation was added, and the configuration is constant independently of the toroidal angle ϕ. To introduce the perturbation of the non-axisymmetry of the coils, an additional non-axisymmetric $\delta\psi$ contribution is computed. This is described in the next chapter.

6.3 PF coil contributions and non-axisymmetric perturbation

In this chapter, the calculation of the error-field caused by the non-axisymmetry of the coil location is described. As shown in sec.5, the poloidal field coils are not perfectly axisymmetric. Since the confined plasma equilibrium reconstruction of LIUQE relies on perfect axisymmetry, the small displacements create a field perturbation that disrupts the perfect confinement properties of a purely axisymmetric plasma configuration. The poloidal flux function $\psi(R, Z)$ in an axi-symmetric tokamak equilibrium is described as the sum of two contributions, as shown in eq.(25). The perturbation, which will be referred to as $\delta\psi$, can be approximated using a first-order expansion of the coil flux contribution ψ_C, similar to the calculation of the coil displacement in sec.5.3. Here, the green function G is used instead of the matrix $\underline{\underline{\mathbf{M_{xc}}}}$ for clarity. The summation is also explicitly provided, rather than relying on the Einstein convention.

As expressed by eq.(26), $\psi_C(R, Z)$ is the sum of the poloidal flux created by each poloidal field coil with radius and height (R_c, Z_c) at a certain location in the section with cylindrical coordinates (R, Z). To account for perturbations caused by the small coil geometry discrepancies, a first order expansion is applied to the Green's function of eq.(26). For a small ΔZ_c displacement, the expansion is performed with respect to the Z_c component of the coil in the green's function: $G(R_c, Z_c + \Delta Z_c, R, Z) \simeq G(R_c, Z_c, R, Z) + \Delta Z_c \frac{\partial G(R_c, Z_c, R, Z)}{\partial Z_c}$. The ψ_C perturbation caused by the displacements of the radius and the height location of each coils is given by the following expression:

$$\begin{aligned} &\psi_C(R, Z) + \delta\psi(R, Z) = \\ \sum_{i=1}^{N_c} I_i \Bigg(G(R_i, Z_i, R, Z) &+ \Delta R_i \frac{\partial G(R_i, Z_i, R, Z)}{\partial R_i} + \Delta Z_i \frac{\partial G(R_i, Z_i, R, Z)}{\partial Z_i} \Bigg) \ . \end{aligned} \tag{27}$$

The R_c and Z_c displacements are, however, axisymmetric. They should be contributing to the LIUQE reconstruction as it includes the magnetic probes measurements in its algorithm, which capture the poloidal field. Regardless, this type of perturbation does not create any chaos or turnstile lobe, as they are axisymmetric. As mentioned before, only the tilt $\Delta\alpha^{ox}$, $\Delta\alpha^{oy}$, and the cartesian displacement of the center of the coil ΔX_c, ΔY_c create a $n = 1$ perturbation mode. The whole expression is equivalent to eq.(14), but with the flux function $\psi(R, Z)$ instead of the saddle loop flux ψ_s.

For tilts, the equations are developed using the same assumptions as in section 5.3 and follow the same chain rule development. $\frac{\partial G(R_c, Z_c, R, Z)}{\partial \alpha_{cy}} \Delta\alpha_{cy} \simeq -\frac{\partial G(R_c, Z_c, R, Z)}{\partial Z} \cos(\phi) \Delta_t Z_c^x$. The variable Z_c^x is the maximum height displacement caused by an α_{oy} tilt, which is localized on the X axis. The derivative can be performed directly on the Z_c component since $-\partial Z_c = \partial Z$. For consistency, the derivation will also be conducted on the coordinate component Z. The development is similar for ox tilt and the full expression

$\delta\psi_{\Delta\alpha^{ox,oy}}$ becomes:

$$\delta\psi_{\Delta\alpha^{ox,oy}}(R,\phi,Z) = -\sum_{c=1}^{N_c} I_c \frac{\partial G(R_c,Z_c,R,Z)}{\partial Z} \left(\Delta_t Z_c^x \cos(\phi) + \Delta Z_c^y \sin(\phi)\right) \ . \tag{28}$$

A new ϕ-dependence is now added and the axisymmetry is broken. For the displacement of the center of the coil ΔX_c and ΔY_c, the same assumptions as in section 5.3 can be used. A displacement of ΔR_c^{center} of the center of the coil is considered to be a displacement of $-\Delta R$ in the coordinate system of (R, Z). Using the chain rule, the local displacement R of the coordinate in a ϕ section is then given by $\Delta R = \Delta X \cos(\phi) + \Delta Y \sin(\phi)$. The equation for the perturbed $\delta\psi_c$ due to the shift of the coil center then becomes:

$$\delta\psi_{\Delta X,Y}(R,\phi,Z) = -\sum_{i=1}^{N_c} I_c \frac{\partial G(R_c,Z_c,R,Z)}{\partial R} \left(\Delta X_c \cos(\phi) + \Delta Y_c \sin(\phi)\right) \ . \tag{29}$$

The complete expression utilized in the code, organized in a matrix and separated into a cosine and a sine component, is expressed by the three following expressions:

$$\psi(R,\phi,Z) = \psi(R,Z) + \delta\psi_{cos}(R,\phi,Z) + \delta\psi_{sin}(R,\phi,Z) \ , \tag{30}$$

$$\delta\psi_{cos}(R,\phi,Z) = -\left(\frac{\partial \underline{\underline{\mathbf{M_{xc}}}}}{\partial R}\Delta X_c I_c + \frac{\partial \underline{\underline{\mathbf{M_{xc}}}}}{\partial Z}\Delta Z_c^{oy} I_c\right)\cos(\phi) \ , \tag{31}$$

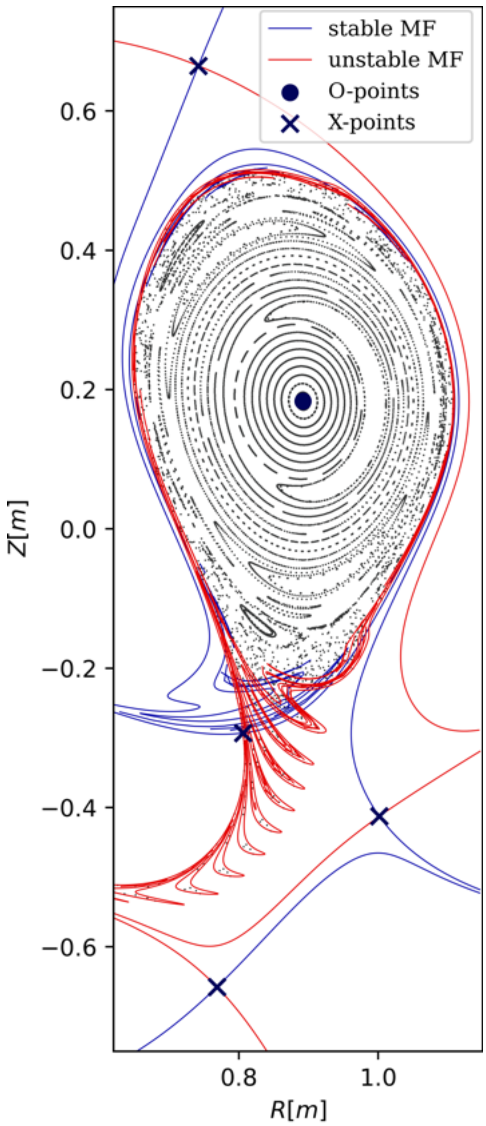
$$\delta\psi_{sin}(R,\phi,Z) = -\left(\frac{\partial \underline{\underline{\mathbf{M_{xc}}}}}{\partial R}\Delta Y_c I_c + \frac{\partial \underline{\underline{\mathbf{M_{xc}}}}}{\partial Z}\Delta Z_c^{ox} I_c\right)\sin(\phi) \ . \tag{32}$$

Now the complete ψ function has an explicit ϕ component, which means that the field is no longer axisymmetric and the amplitude of the perturbation varies toroidally. The grid containing both $\delta\psi_{cos}$ and $\delta\psi_{sin}$ (without the $\cos\theta$ and $\sin\phi$ terms) is computed in Matlab using the displacement that we calculated in sec.5.3. The grid is identical to the axisymmetric ψ grid. They are computed for each specific shot, as the current I_c depends on the shot and the time. The grid with the $\delta\psi$ is then used in python as an input for a `pyoculus` instance that integrates perturbed field lines based on a discrete grid.

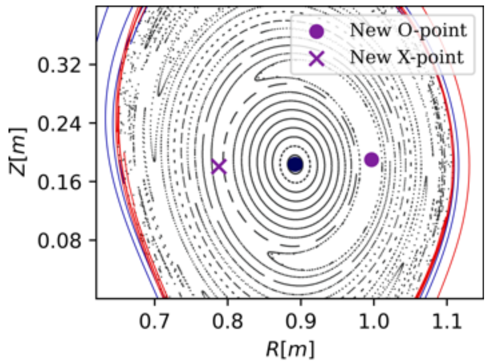
The Poincaré plot computed for the same shot as fig.18 (75979) is shown in fig.19a. The broken separatrix now allows transport across the confined region through the turnstile lobes, as described in sec.4.3. Field lines in the turnstile lobe of the exiting set just at the top of the separatrix X-point, represented by the blue area in fig.19c, rotate clockwise around the LCFS, as they are iterated forward and finish in the red stripe lobes after 8 iterations, represented by the red area. The turnstile flux is computed using the Meiss formula, expressed by eq.(40), in sec.A.2. They are pushed to the high field side as iterations increase, and the magnetic field lines finish their journey in plasma facing components.

The perfect nested constant surfaces are now broken in some regions. They are replaced by magnetic island chains surrounded by stochastic regions. The initial fixed-point displacements are small; however, new fixed-points have been created. The creation of a magnetic island chain is associated with the formation of pairs of O-points and X-points. Here, a single island to the right of the magnetic axis, with an O-point and X-point represented in purple, is illustrated in fig.19b.

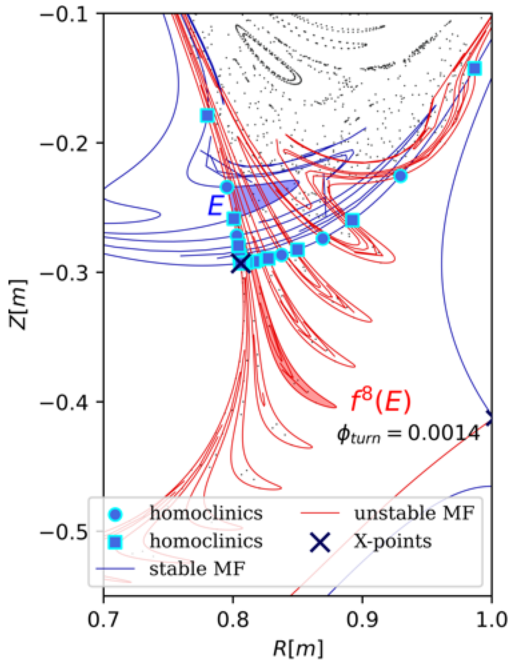
Their patterns already show a significant resemblance to the image from the MANTIS camera, as shown in sec.2. To compare the shapes of the emissivity poloidal profile of MANTIS with the patterns of the manifolds, a MANTIS/ manifold plot will be utilized. This will be the subject of the next chapter. Both diagnostics will be compared, and the discrepancies between them will be analyzed. The goal is to provide evidence that the Jellyfish's tentacles are the turnstile lobes of the error-field induced homoclinic tangle.



(a) Whole section



(b) Island chain, creation of a pair of fixed-point



(c) Magnetic tangle and turnstile lobes

Figure 19: Poincaré plot for shot 75979, at 1.2s, with error field induced perturbation. Turnstile lobes and magnetic island have appeared

7 MANTIS comparison

Last chapter, we saw that introducing the error-field perturbation in TCV shots lead to the creation of strange patterns in the divertor area. These patterns are called turnstile lobes. Within these lobes reside magnetic field lines that connect the inner region to the exterior. Particles within the plasma may subsequently end up in these lobes, resulting in their transport outside the confinement region. These were the main suspect of being the jellyfish's tentacles, visible in the MANTIS cameras.

As described in sec.2, the MANTIS diagnostic provides images of line radiation from different atomic transitions. By applying tomographic inversion techniques, measurements allow the reconstruction of the local emissivity distribution in a poloidal cross-section of the divertor region. Emissions result from by the radiative de-excitation of excited atomic or ionic states. These excitated states are primarily caused by electron-impact and electron recombination excitations. The emissivity depends on the local electron density n_e, temperature T_e, and ion density n_i, and is described by the CRM model, expressed in eq.(1). Region of large emissivity generally corresponds to high densities locations combined with favorable temperatures to excitations of a specific transition. Plasma originating from the confined zone has a relatively high T_e and particle densities $n_{i,e}$.

As turnstile transports connect field lines from the inside to the divertor region, it contains particles from the confinement with favorable densities for line emission. Even if non-Hamiltonian processes, such as perpendicular transport, is neglected in field-line tracing models[12], enhanced emissivities are expected inside the turnstile lobes but not guaranteed. It depends on the temperature, too hot or too cold and the specific spectral line properties. The MANTIS diagnostic can, therefore, be used as an experimental tool to investigate correlations between emissivity patterns and the location of the turnstile lobes.

In this chapter, an analysis of Poincaré simulations and MANTIS tomographic reconstruction images, is conducted. The goal is to provide evidence regarding the assertion that the turnstile lobes are, in fact, the tentacles of the jellyfish. In the first part, a quick description about the numerical routine will be discussed, along with a first qualitative comparison on the first result. Secondly, the comparison of turnstile patterns will be extended to include more shots, specifically those exhibiting distinct strange patterns and those without any visible patterns. Then, the reliability of the MANTIS tomographic reconstruction images and the comparison among different spectral lines will be examined. Finally, we will be focusing on one specific tentacle pattern property, which is the inter leg distance.

7.1 First result and routine description

To compare patterns of the turnstile lobe computed in simulations results with experimental images observed in TCV, the manifolds computed with `pyoculus` are plotted on top of the MANTIS image.[2]. To execute that, the triangular inverted grid of MANTIS is extracted in matlab and used in python, where Poincaré simulations are computed. The manifolds and the fixed-points are superposed with the colormap of the relative emissivity in MANTIS. Calibrated images are not required in this simple qualitative visual analysis, as the patterns are already sufficient to make correlations with turnstile lobe locations. In this section, a first analysis is conducted on the shot 75979.

In fig.20, a MANTIS/manifold comparison is shown for the Jellyfish discharge 75979. In fig.20a, the full TCV poloidal cross-section is displayed, including the stable and unstable manifolds, X-points and O-point, together with the emissivity colormap from MANTIS. The divertor (unstable) manifold is highlighted in white. This manifold connects the X-point separatrix to the HFS divertor target. Particles located close to the X-points are expected to follow magnetic field lines near this manifold, which connect to the divertor target. The Poincaré section is computed at the $\phi = 1.9$ rad toroidal section, which is slightly less than $5\pi/8$ rad. This angle is estimated to correspond to the MANTIS viewing angle, as deduced from the diagnostic setup representation[36]. The spectral line used by MANTIS is a He-II line. Yellow regions indicate high emissivity at the corresponding wavelength. In the following, these regions are assumed to be indicative of locally high particle densities, for both electrons and ions.

In fig.20b, images of the divertor region are presented with manifolds, and in fig.20c, without, to provide a clear representation of the emissivity distribution. In both diagnostics, high emissivity patterns in MANTIS and the red manifolds for the turnstile lobe, stripe patterns are appearing in the divertor region. The similarity between them is undeniable. The qualitative analysis is based on two keypoints: the shape of the tentacles and their locations.

First of all, the shapes of the patterns show significant similarities. The MANTIS high emissivity stripes and lobes in red, depart from the divertor manifold in the HFS and point towards the LFS. In both representations, the patterns demonstrate a similar curvature, which seems to be directed towards one of the secondary X-points. The stripes in MANTIS however, are longer than the red lobes and thicker. They could be caused by discrepancies in the displacement calculation. However, precise match is less expected, as perpendicular diffusion process are neglected in field-line tracing simulations.

The location of the tentacles is matching relatively well, specifically for the three legs whose tips lie approximately between $Z \in [-0.45, -0.38]$. The MANTIS angle of view seems to be correctly estimated. For the lower legs, there appears to be a growing dephasing as the lobes are iterated forward. The "spatial periodicity" of the legs seems to be higher in the MANTIS diagnostic than in the magnetic tangle. Due to this observation, one additional bottom leg is visible in MANTIS. The divertor leg, is also matching with a

[2]To simplify, the MANTIS tomographic inversion reconstruction image are just referred to as MANTIS image

high emissivity stripe. It indicates the region for which particles are guided along the field line to the divertor target. The "V-shape artifact" is particularly visible in this image, it is the triangle that departs from the LFS at $Z \simeq -0.45$ and extends horizontally towards the HFS, as shown in fig.20a. The emissivity profile reconstruction within this region is degraded.

In this case, the similarity in shape and spatial location between the turnstile lobes and the MANTIS image provides convincing evidence of their correlation. To maximize evidence, analyzes are conducted on alternative plasma configurations in the next section. MANTIS/manifold plots will be displayed in shots where long turnstile lobes directed toward the LFS are expected and where they are not, relative to the MANTIS images.

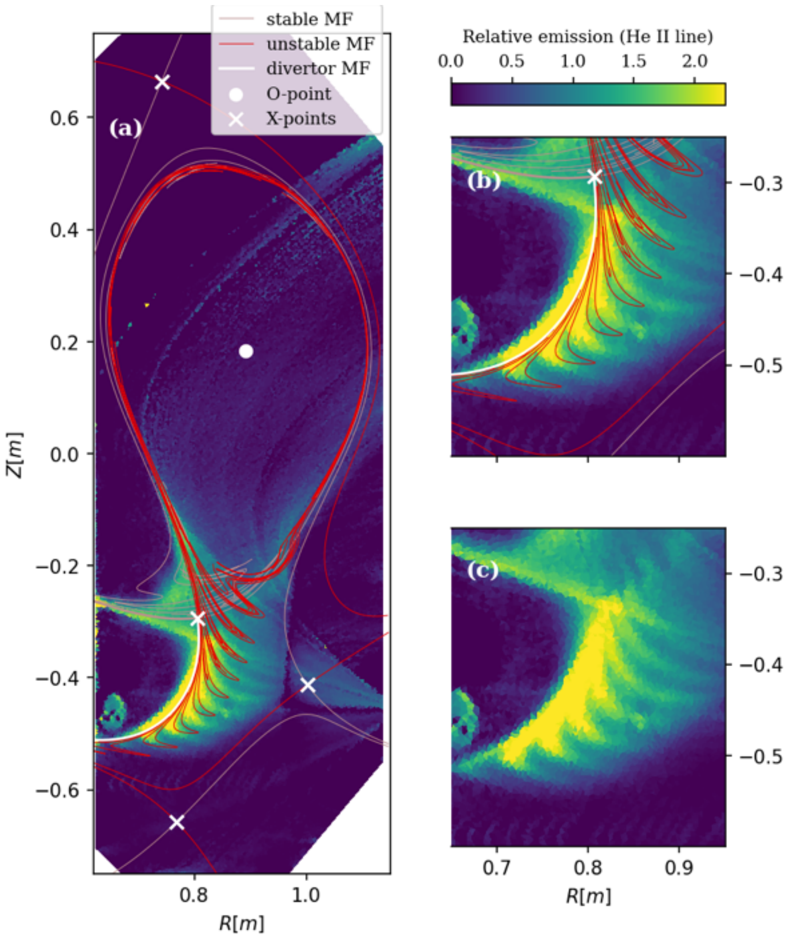


Figure 20: Overplot of MANTIS tomogaphic reconstruction for He-II line and error-field manifolds/ fixed point for shot 75979 at 1.2 s.

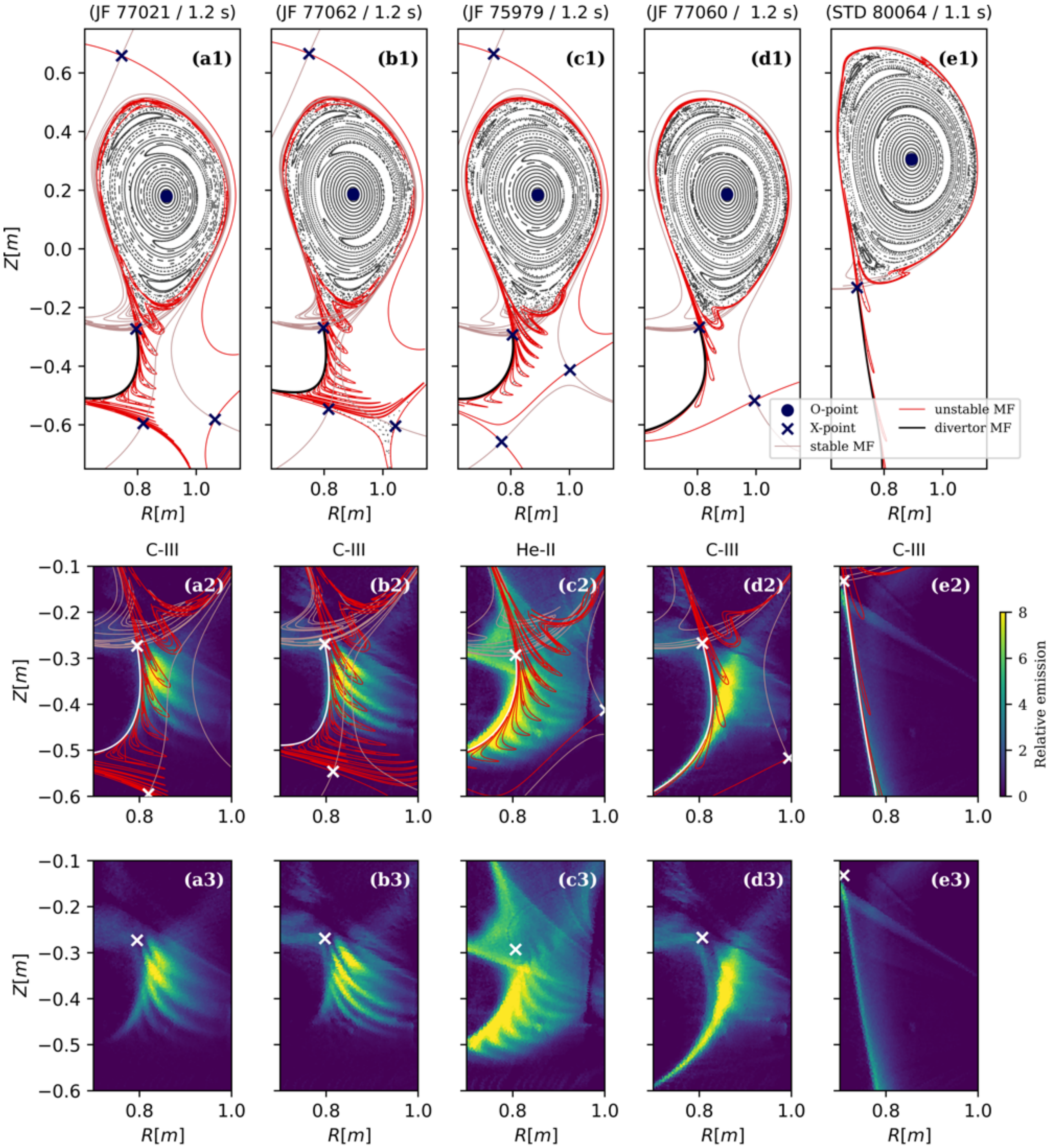


Figure 21: Error-field tangles for 4 Jellyfish shots and one standard shot. On top, poincaré plots with manifolds/ fixed-points. On the middle, MANTIS/ Manifold plot zoomed in the divertor region. On bottom, simple MANTIS diagnostic plots. The appearance of the tentacles seem to correlate with the manifolds.

7.2 Turnstile lobes qualitative analysis

As shown in the previous section, clear evidences based on pattern similarity was provided to demonstrate the correlation between the tentacles in MANTIS and the turnstile lobes in one shot. In this section, 4 other configurations will be simulated: two JFs with visible tentacles in the MANTIS image (fig.21a / fig.21b), one JF without tentacles[3] (fig.21d) and one standard single-null configuration (fig.21e). Their timing was either chosen arbitrarily or based on the quality and clarity of the tentacles in the MANTIS image. The line used is either the C-III line or He-II, which provided the least polluted images and clearer representations of the tentacles.

In fig.21a2 and fig.21b2, where MANTIS shows stripe patterns of high emissivity, elongated turnstile lobe patterns appear. Like shot 75979, the lobes depart from the divertor leg and are directed towards the secondary X-point, exhibiting a similar curvature. The location of the LFS X-point in a1 and b1 is different from that in c1, as it is situated approximately $10[cm]$ below. It alters the structure of the manifolds and the direction where the turnstile lobe point towards. Overall, the correlation MANTIS/manifold based on tentacle shapes are convincing for 77021/ 77062.

Regarding the location of the legs, the results show a significant discrepancy for shots 77021 and 77062. In fig.21a2 and fig.21b2, the first lobe (closest to the X-point) does not match with MANTIS, as it is located between two large emissivity stripes. This does not seem to be a phase problem, as the MANTIS emissivity correlates with the second and third turnstile lobe (from the top X-point). There is an additional top leg in MANTIS that may also be attributable to an underestimated spatial periodicity of the turnstile lobe. The spatial location shows less conclusive evidence in 77021/ 77062 than in 75979. The discrepancy of the first lobe might be a feature of the C-III line tomographic inversion, which will be discussed in detail in sec.7.3.

Regarding the representation of the divertor manifold, there isn't a clear emissivity line covering the whole manifold, from the X-point to the HFS target. It is unclear whether the computed X-point is mislocated or if it is related to the properties of the C-III line tomographic reconstruction.

The other two tentacled shots also provide solid MANTIS/ manifolds correlation, but with more discrepancies than shot 75979. What about the shots without visible tentacles ? In fig.21d and in fig.21e, two shots are displayed for which no clear turnstile lobe patterns similar to tentacles are expected.

In fig.21d2, turnstile lobes are still present but are shorter, with only three lobes along the divertor manifold. One significant difference is that they curve in the clockwise direction, contrary to the previous 3 shots, which curved anticlockwise, and they are significantly less spread out towards the LFS. They stay close to the divertor leg and MANTIS does not capture tentacle patterns. Even in the absence of a clear match with the MANTIS image, their patterns seem to correlate with the prominent emissivity region in the divertor area. In this shot, the same observation regarding the coverage

[3]This is in fact not a proper Jellyfish configuration, as there aren't 3 X-points. This would be a Snowflake minus.

of emissivity along the divertor manifold is noted. The extra X-point in the 3 JFs configuration has an effect on the turnstile lobe, it causes them to point away from the divertor manifold.

For the standard shot displayed in fig.21e2, turnstile lobes are visible, but they are located very close to the divertor manifold. It correlates with MANTIS, which shows only light along the divertor manifold.

In this section, we demonstrated that MANTIS reveals tentacle patterns in shots where turnstile lobes depart from the divertor manifold and elongate towards the HFS. In instances where MANTIS does not exhibit these patterns, the turnstile lobes remain considerably closer to the divertor manifold. However, there are still a few discrepancies. The distance between the legs and their lengths do not correlate completely. Moreover, the number of tentacles does not match in the X-point region for shots 77021 and 77062, where the C-III line is used.

Another observation is that MANTIS shows a lack of emissivity along the divertor manifold, specifically in shots with the C-III line. There are no sign in MANTIS that the X-point location is correctly computed in the field-line tracing simulation. This also raises questions regarding the reliability of the C-III spectral line in representing the density profile in the divertor region, given that the best spatial correlation was observed in shot 75979 with the He-II spectral line.

In the following section, answers will be provided based on the spectral line analysis.

7.3 Spectral line comparison

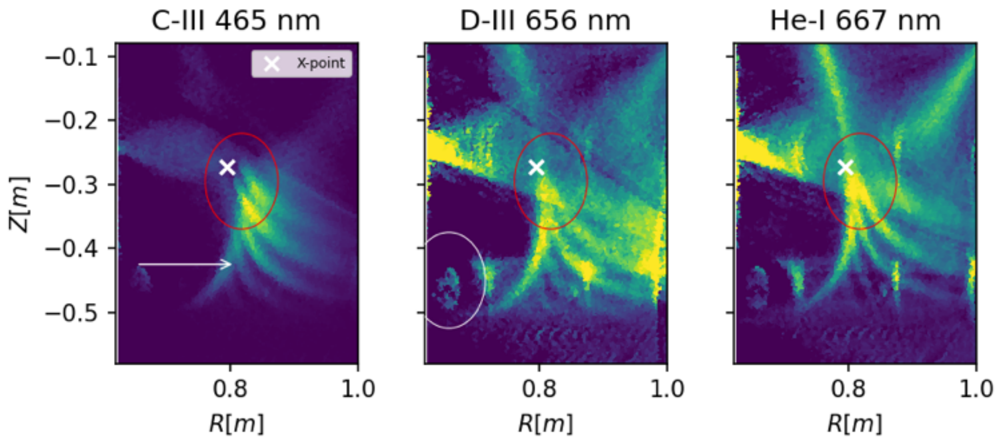


Figure 22: MANTIS Tomographic reconstruction for shot 77021 at 1.2 s. with 3 different spectral line.

As seen in previous section, there are a few discrepancies between what is seen in MANTIS and the shape of the manifolds. As shown in fig.21a2 and fig.21b2, the number of legs do not match. There is what seems to be an extra top tentacle visible in MANTIS, where the C-III line is used. This specific line was employed primarily in this work because the tomographic reconstruction demonstrated the highest image quality. However, a recent visual analysis of fig.23 has called into question the profile computed with the C-III line tomographic reconstruction.

In fig.23, there is a raw MANTIS image of shot 77021 at 1.2s for the C-III line, corresponding to the tomographic reconstruction in fig.21a. By looking at the height location of the clear spot in the bottom right (corresponding to XTOMO/ bolo/AXUV diagnostic), the last visible tentacle (red arrow) could be matched with the one in the tomographic reconstruction. In fig.22, three different line tomographic reconstructions are plotted, and the same clear spot, surrounded by a white circle, appears. The same last visible raw MANTIS data tentacle, indicated by the red arrow in fig.23, is identified and marked by a white arrow in fig.22. The observation was that there are only three visible tentacles in the C-III raw MANTIS data. This suggests that the tentacle surrounded by a red circle in fig.22/ C-III, could be an artifact created by the tomographic inversion.

Moreover, in the He-I line and the D-III images, what seem to be an extra tentacle is no longer here, as high emissivity stripes connect to the X-point. There are also clear high emissivity regions along the divertor manifold.

The source of this artifact could come from different reasons. As can be seen, the C-III line is not emitting near the X-point. As explained in sec.6, C-III, which is an impurity, can go abruptly from an emitting state to a non-emitting state depending on the density and temperature of the plasma [5]. When it is too hot, carbon is unlikely to be present in C^{2+} (for EX) or in C^{3+}(for REC) forms. It is not emitting in the last closed field surface and around the X-point, in contrast to He-I and D-III. Combined with the V-shaped artifact [5], which degrade the emissivity profile, it could give the impression of an extra tentacle shape.

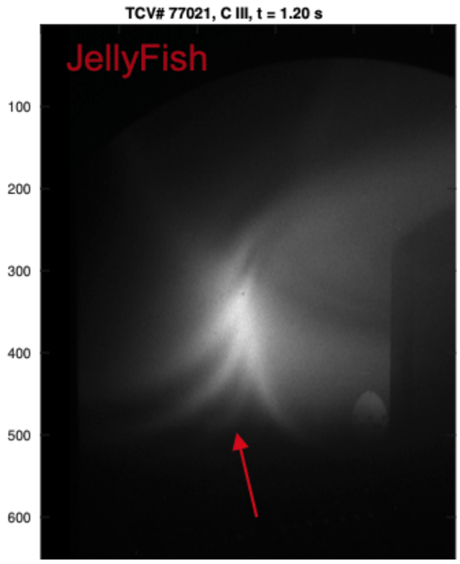


Figure 23: Raw data from the MANTIS diagnostic, taken from A.Perek presentation [37]

Even if the signal is more polluted than C-III, the helium line seems to provide a better 2-D density map, as shown in the MANTIS/manifold plot in fig.24. This shows how careful tomographic reconstruction images must be treated. The result for 77021 is now much more convincing. The top tentacle, the X-point and the divertor manifold locations correlate strongly with the MANTIS emissivity profile.

It should also explain in fig.21b2, the extra leg in shot 77062 and the lack of emission

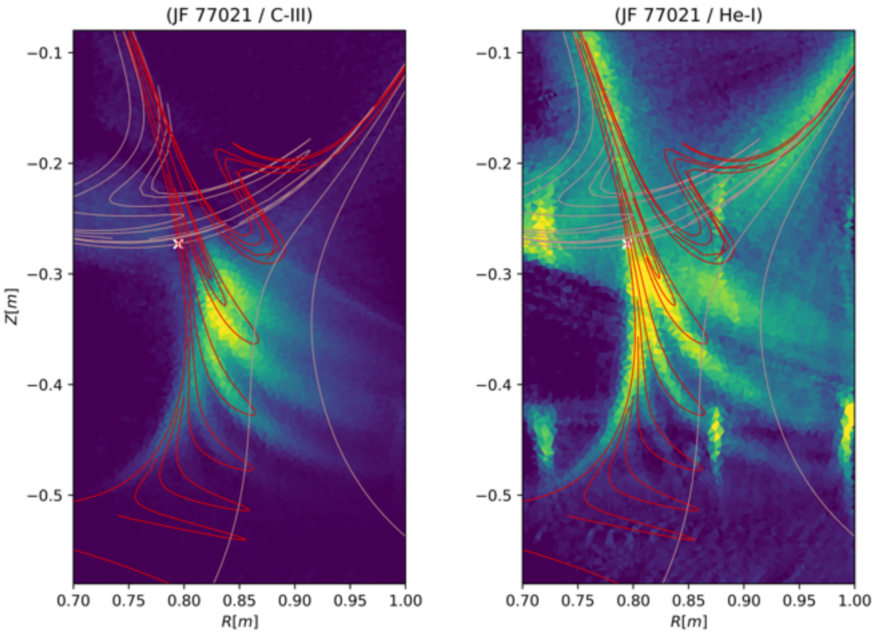


Figure 24: C-III and He-I tomographic inversion for shot 77021 at 1.2 s. The stable and the unstable manifolds are overplotted on top.

along the divertor manifold for every C-III line MANTIS image. Another tomographic reconstruction line for 77062 was computed during the last few days before the deadline of this thesis. Discussion in section 7.2 will be kept as it is due to the lack of time. For the next section, the new tomographic reconstruction with He-I (706 [nm]) for 77062 will be used. As expected, the extra leg is no longer present, as shown in fig.26.

Even if the C-III tomographic reconstruction resolves a few of our concerns, there are still discrepancies. The lengths of the tentacles are different, and there is still a discrepancy in the tentacle periodicity between what MANTIS shows and the turnstile lobes, which will be discussed in the next section.

7.4 Average distance between legs

Previously, distance between lobes was found to be different in the MANTIS image and the magnetic tangle. In this section, this will be quantify by computing the average distance between the leg and will be link to a difference in poloidal field.

To compare the location of the tentacles in the MANTIS diagnostic with the turnstiles, a 1-D line is plotted on the divertor region. The emission along this line and the intersection with the manifolds are computed. The location of the tentacle will be associated with the peaks of the emissivity, represented by blue dotted lines and blue dots, as is shown in fig.25 for shot 77021. A gaussian fit is added to facilitate the process [38]. The fit has to be visually checked and the standard deviation adjusted to have a consistent match between the fit and the raw emissivity. For the manifolds, the location of the tentacle is assumed to be the middle point between the intersections of the line and the corresponding manifolds, represented by a dotted red line.

The average distance between the legs is computed for MANTIS (Δd_{emi}) and the turnstile lobes (Δd_{MF}), shown in fig.25, fig.26 and fig.27. The ratio between them and the approximated difference are shown in table 1. It is used as rough estimation for quantifying the lobe spacing.

For shot 77021, shown in fig.25, the red line and the blue line are well matched for the second and third legs. For the first leg, the turnstile lobe is shifted upward and for the fourth leg, it is shifted downward. This can be associated to a smaller spatial periodicity of the legs. The ratio between them avg Δd is $\frac{\Delta d_{emi}}{\Delta d_{MF}} \simeq 0.83$.

For shot 77062, shown in fig.26, the new spectral line gives a better spacial location but is more polluted. The ratio between their avg Δd is $\frac{\Delta d_{emi}}{\Delta d_{MF}} \simeq 0.91$.

For shot 75979, shown in fig.27, the turnstile lobe are shifted downward. The ratio between their avg Δd is $\frac{\Delta d_{emi}}{\Delta d_{MF}} \simeq 0.87$.

The conclusion is the same for the 3 shots. The tentacles are closer in the MANTIS tomographic reconstruction. Even if the plots were only shown for 3 shots at one instant, other simulations at different timings showed similar conclusion.

estimated values	77021	77062	75979
$\gamma = \frac{\Delta d_{emi}}{\Delta d_{MF}}$	0.83	0.91	0.87
$(\Delta d_{MF} - \Delta d_{emi})[mm]$	9.2	4.4	5.3
$\delta B^Z[mT]$	2.7	1.3	1.6

Table 1: discrepancies of inter leg distance ratio, difference and estimated poloidal field uncertainty for three jellyfishs

This could actually be a sign of an uncertainty in the poloidal field. In order to estimate the order of magnitude of the observed discrepancy, we introduce a deliberately simplified model. The following assumptions are made: only the vertical component B^Z is retained for the poloidal field, B^ϕ and B^Z are assumed constant over one toroidal turn, and radial field is negelected.

Under these assumptions, the vertical displacement after one toroidal turn can be approximated using the Poincaré map integration as

$$Z_2 \simeq Z_1 + 2\pi \frac{B^Z}{B^\phi}, \qquad \Delta Z = 2\pi \frac{B^Z}{B^\phi}. \tag{33}$$

This expression provides an order-of-magnitude estimate of the vertical displacement. For the three shots, the ϕ component of the magnetic field is $B_\phi \simeq 1.5[T]$.

An uncertainty of $\delta B^Z \sim 1$ mT induces, within the present simplified model, a vertical shift $Z_{\text{shift}} = \Delta Z_b - \Delta Z_a = 2\pi \frac{\delta B^Z}{B^\phi} \sim 3.4$ mm. The differences between the average inter-leg distances are in the 0.4-1 cm range. The corresponding values of δB^Z required to produce shifts of comparable magnitude are reported in tab.1. They are in the mT range.

Standard deviations of about 20% are observed for discharges 77062 and 77021, reflecting variations of the lobe spacing along the one-dimensional sampling line. This calculation is therefore only intended to provide an order-of-magnitude estimate of the poloidal field variation.

First of all, it means that uncertainties in the order of the mT in the divertor region could match the order of magnitude of the discrepancies in the average leg distance, which is small.

Secondly, it shows how manifolds are sensitive to magnetic field uncertainties. The poloidal field uncertainty of this order could come from magnetic measurement diagnostic uncertainties. If magnetic probes, for example, are badly calibrated, inducing uncertainties on the mT, their use in the LIUQE reconstruction equilibrium reconstruction could have effect on the poloidal field on the same order of magnitude. Another source which could modify the poloidal field component is the presence of the scrape-off layer current in the divertor's manifold. This will be discussed in a next section.

In this section, discrepancies observed in the spatial periodicity were quantified and linked to uncertainties in poloidal fields. It gave us a good grasp on the manifolds sensitivities to poloidal fields. In the next chapter, more explorations are going to be be conducted on other potential sources of uncertainty.

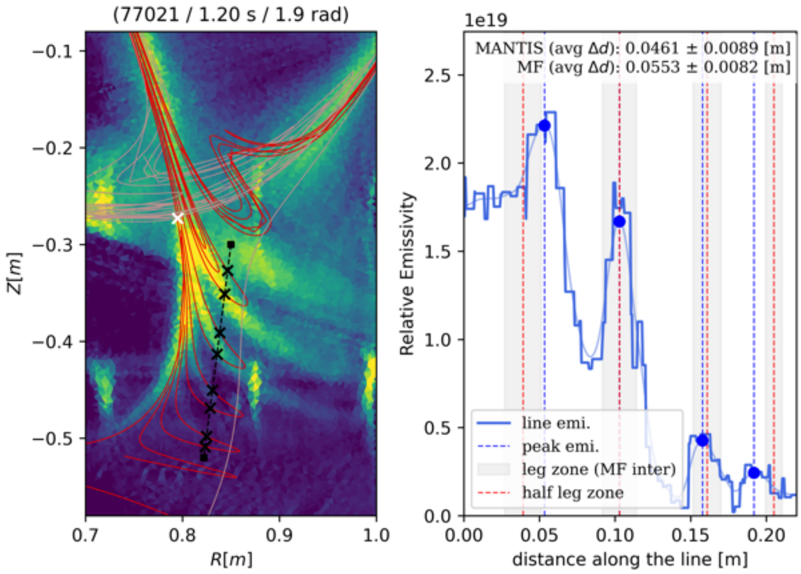


Figure 25: Tentacle spacing analysis plot for shot 77021, with He-I (667 [nm]) line tomographic reconstruction.

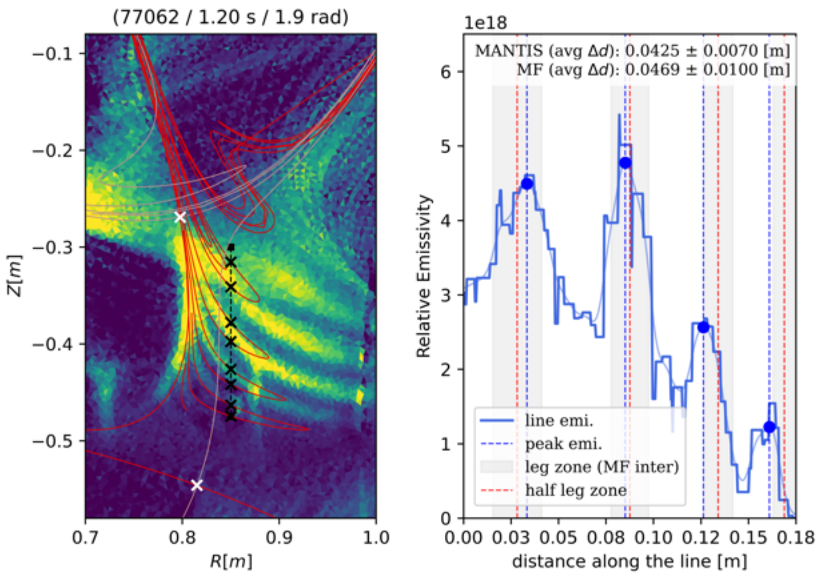


Figure 26: Tentacle spacing analysis plot for shot 77062, with He-I (706 [nm]) line tomographic reconstruction.

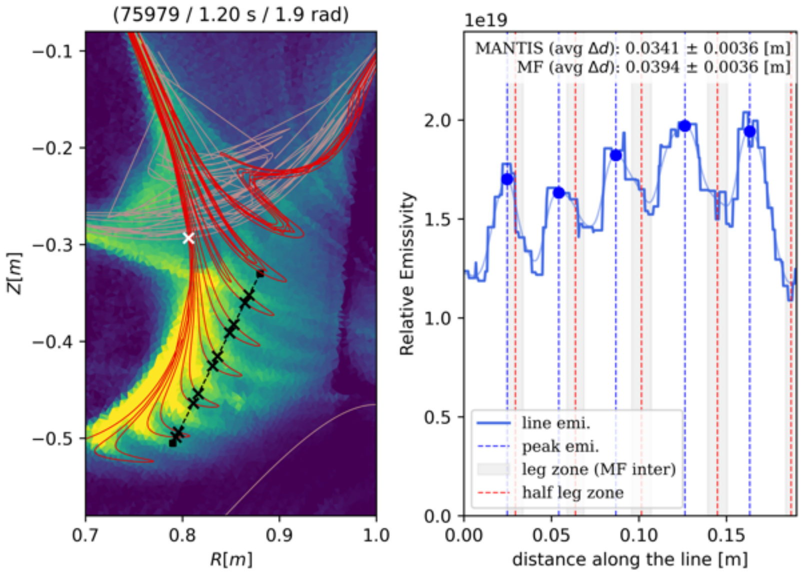


Figure 27: Tentacle spacing analysis plot for shot 75979, with He-II (468 [nm]) line tomographic reconstruction.

8 Explorations on tentacle patterns discrepancies

As field line mapping only takes into account the topology of the magnetic field, many physical processes are neglected. For this reason, the turnstile lobes are unlikely to exactly match the image provided by the tomographic reconstruction. Moreover, the MANTIS diagnostic also has inaccuracies in tomographic inversion that serve as sources of artifact creation. The assumption of the perfect toroidal symmetry in MANTIS is contradicted by the error field tangle, as it creates non-axisymmetry.

However, the most significant emissivity captured by the camera lens can be presumed to be concentrated in sections that are in the vicinity of the perpendicular section relative to the lens field of view. In this region, the locations of the turnstile lobes do not differ significantly enough to contribute to a misplacement of the tentacles in MANTIS. The convolution of non-axisymmetry in an axi-symmetric reconstruction is likely a contributor to the thickening and blurring of the tentacles.

Therefore, explorations will be conducted on the poincaré map aspect of the comparison. The first exploration will be conducted is a synthetic analysis of the effects of tilts and displacements on the manifold patterns. The goal is to observe how these factors affect the length of the legs and their periodicity.
Secondly, the effect of a divertor current leg will be analyzed. The goal is to determine whether the order of magnitude of the magnetic field resulting from it could match the discrepancy of the average peak distance.
In the third and final section of this study, several adaptations of the LIUQE model will be made to evaluate how the patterns of the turnstile lobes impact the results. The kinetic model will be utilized, and a manual adjustment of the plasma current I_p will be made in the LIUQE model.

8.1 Synthetic analysis of displacement

To evaluate the effect of various types of displacement, different coil configurations are generated to compute manifolds. Since there are 18 coils with 4 degrees of freedom, the analysis is restricted to the E coils. These coils exhibit the largest displacement and error bars, as shown in sec.5. They are primarily responsible for generating the magnetic tangles. A coil configuration test conducted with the isolation of each set of coils yields consistent conclusions. To further limit the numerous configurations of coils, the axisymmetry of a coil and the characteristic of the magnetic tangles are utilized. For any displacement with a constant norm, changing the angle will only modify the phase of the magnetic perturbation, and similarly for tilt.

Two non-combined configurations are used, $\Delta\alpha^{oy}$ and ΔX. The total displacement is increased and set at $\Delta_t Z^{oy} = \Delta X = 2[cm]$ for every E coil. For combining tilts and displacements, only the coil displacement relative to its tilt is relevant. Three combinations are used: $\Delta\alpha^{oy}/\Delta X+$, $\Delta\alpha^{oy}/\Delta X-$, and $\Delta\alpha^{oy}/\Delta Y+$.

Secondly, displacement configurations are generated by adding a $\Delta X+$ only for certain E coils. The shot used in this section is shot 77021 at 1.2 seconds. The manifolds are then computed and compared to the original, with the displacement coming from the

LSE fitting.

8.1.1 Tilt and displacement comparison

As shown in fig.29a/b, $\Delta X+$ and $\Delta\alpha^{oy}$ generate similar patterns of manifolds. The lengths of the turnstile lobes do not differ significantly.

The turnstiles do not have the same phase, and they are slightly thicker for the tilts. A magnetic perturbation distribution plot is shown in fig.30 to visualize the difference. As can be seen, the B_R distribution for $\Delta X+$ is equivalent to the B_Z tilt distribution but with a π phase shift. Due to the linearized green functions, their expressions are similar; as the displacements are equal, they generate similar field perturbations. The fact that the geometries are nearly equal is quite surprising.

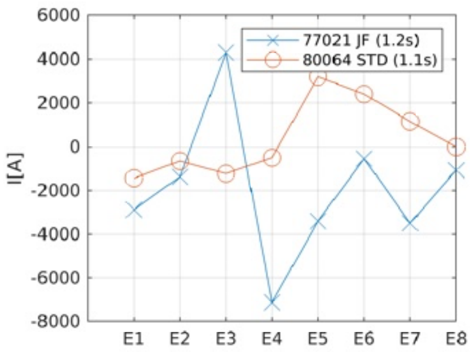


Figure 28: E coil current for the Jellyfish shot 77021 and standard shot 80064.

For $B_Z/\Delta X$ and $B_R/\Delta\alpha^{oy}$, the results are the same, with the only difference being that the norm is slightly smaller for $B_Z/\Delta X$. With this plot, the locations of certain coils and their impacts can be identified. The E3 and E4 coils generate significant perturbations. As shown in fig.28, they exhibit a larger current than the other E coil, $I_{E3} \simeq 4000[A]$ and $I_{E4} \simeq -7000[A]$.

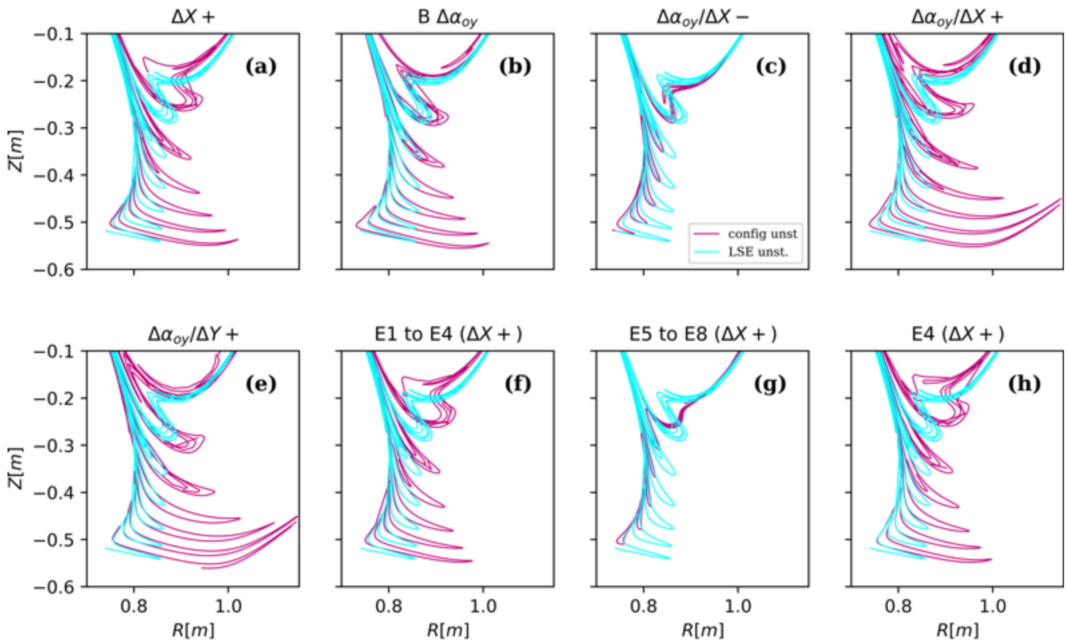
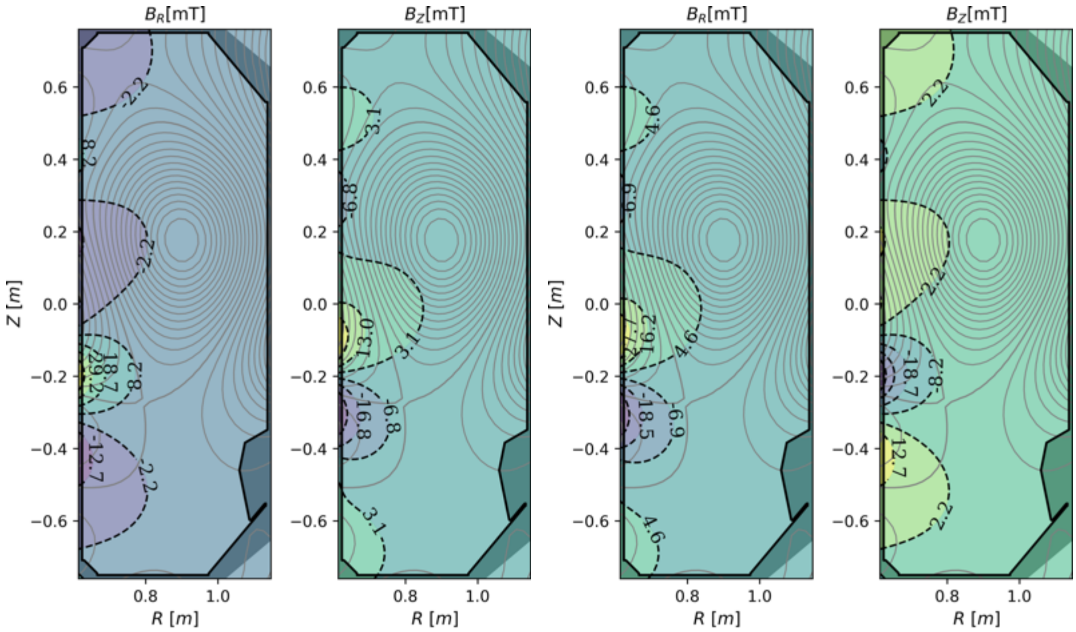


Figure 29: Manifolds computed with different E coil displacement configurations



(a) $\Delta X = 2[cm]$ for every E coils.

(b) $\Delta\alpha^{oy} \simeq 1.86[deg]$ on every E coils

Figure 30: B_R and B_Z magnetic field distribution of the error-field perturbation caused by E coil displacement in left and tilt in right.

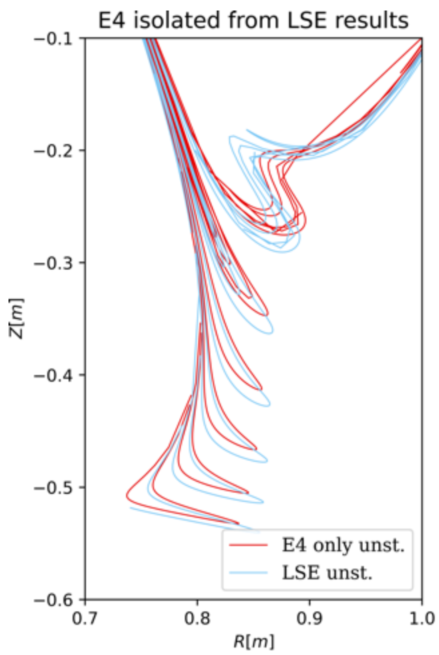


Figure 31: Comparison of the turnstile lobe computed with only the E4 displacements/tilts.

By displacing the E4 coil relative to the vessel at $\phi = 0$, the B_Z component directly in front of the outer surface becomes larger, as the field lines are perpendicular to the outer surface of the coil. This also suggests that the E4 coil has the most significant impact on generating the tentacles of the jellyfish configuration.

As shown in fig.29c/d/e, the combination of tilt and displacement can have different effects. When the coils are displaced in the same direction as the axis where the positive ΔZ of the tilt is located, the perturbations add up. When the displacement is $\pi/2$ shifted, it also adds up, and the turnstile lobes thicken. When it is π shifted, they exhibit a cancelation effect.

The length of the legs depends on the perturbation. It functions similarly to an amplitude factor. However, none of the configurations tested show a reduction in the average distance between the turnstile lobes. The discrepancy discussed in section 7.4 cannot be explained by the uncertainty in the coil displacement calculations.

8.1.2 E coil effect

As suggested in the previous part of the section, the E3 and E4 coils generate the largest perturbations. In fig.29e/f/g, the manifolds were computed with only certain coils displaced. The E5-E8 coils have a minimal effect on generating the manifold structure. By plotting E1-E4 and then only E4, it was shown that the manifolds are primarily generated by the displacement of the E4 coil.

Due to its high activation ($I \simeq 7000[A]$) and location near the confined region, even a small displacement can generate substantial field perturbations. To observe its effect on the configuration with displacements from the LSE, a simulation is conducted using an isolating configuration of E4. This is illustrated in fig.31, where the lengths of the turnstile lobes are nearly similar when isolating only the E4 tilt against all displacements for the E, F, and OH coils.

This section analyzes the effects caused by the different types of non-axisymmetrical displacement on the turnstile patterns. The tilts and displacements have a comparable effect, and their combination contributes constructively or destructively to the expansion of the lobes depending on their relative direction. The effects of separately displacing the E coils were discussed, and it was found that the E4 coil contributes significantly to the turnstile transport. This is due to its large current and location near the X-point. However, no configuration has shown a decrease in the lobe spacing, it mostly contributed to the length of the legs and the phase. In the next chapter, the effect of including a current in the divertor's leg will be analyzed to determine if it affects the lobe spacing.

8.2 Mismatch of the vertical magnetic field

As explained in sec.7.4, there is a mismatch between the average leg distance observed in the MANTIS tomographic reconstruction and in the turnstile lobes. The legs are closer in the MANTIS image. $\gamma = \frac{\Delta d_{emi}}{\Delta d_{MF}} < 1$. We also showed how the poloidal displacement between successive iteration of the Poincaré map are affected by a constant poloidal field, meaning a mismatch could be caused by a discrepancy in the poloidal field. Here the sources which could affect the poloidal field will be discussed, specifically the effect that would be caused by a current in the divertor's manifold. An estimation of the poloidal field, simplified to the vertical field, will be approximated to give an idea of the order of magnitude.

The larger distance observed between the legs for the three discharges could indicate that the B^Z field is underestimated in this region. Changing the displacement did not significantly affect the periodicity of the legs, as observed in sec.8.1. The poloidal field perturbation induced by non-axisymmetric displacements can be expressed as a sum of cosine and sine $n = 1$ periodic components. Since these contributions average to zero over one toroidal turn, increasing the displacement is not expected to modify the poloidal displacement of a field line between iterations, at least in a non-chaotic region.

An $n = 0$ error field perturbation, such as coil radius ΔR_C or height ΔZ_C displacements, could also contribute. However, as such perturbations are captured by the saddle loops and probes used in the LIUQE equilibrium reconstruction, their impact is expected to be minimized. Nevertheless, the various uncertainties in both the equilibrium reconstruction and the MANTIS tomographic reconstruction are expected to cause discrepancies in the results.

A potential cause for this B^Z underestimation is the generation of a magnetic field by the scrape-off layer current. Even though many physical processes are neglected in a field-line tracing model, it is still interesting to estimate the contribution of a possible current flowing in the scrape-off layer along the divertor legs. Given the significant toroidal component of the trajectories in this region, such currents could contribute to the magnetic field generation. A brief description of the scrape-off layer and a first-order estimate will be presented in the next chapter.

8.3 Scrape-off layer current

Surrounding the separatrix, there is a thin plasma layer called the scrape-off layer. This non-confined plasma region consists of expelled plasma particles that follow open field lines directed towards the tokamak wall. In the vicinity of the divertor manifold, the scrape-off layer contains open field lines connecting to the divertor target.

To minimize heat loads on the wall, the connection length $L_{||}$ is increased, such that field lines must complete several toroidal turns before reaching the strike point. Since electrons are much lighter than ions, they are primarily responsible for carrying the current. The field lines located below the X-point separatrix, in the Jellyfishs configuration shown in fig.21, are directed in the positive toroidal direction ($B^\phi > 0$) and in the negative vertical direction ($B^Z < 0$). If electrons follow these field lines, they can generate a magnetic field directed upward in the tentacle region, thereby reducing the vertical displacement between successive toroidal turns. This mechanism could therefore contribute, at least qualitatively, to the discrepancies discussed in sec.7.4.

A detailed description of scrape-off layer physics lies well beyond the scope of this thesis. In the following, a simple approximation of the scrape-off layer current, provided by J.Loizu, is used solely to estimate the order of magnitude of the resulting contribution to B_Z.

The current density flowing parallel to the magnetic field lines is denoted $j_\parallel$ [A/m^2]. In the scrape-off layer, magnetic field lines are predominantly oriented in the toroidal direction in order to increase the connection length and mitigate heat loads on the wall. As a result, the parallel current density can be approximated as $j_\parallel \simeq j_\phi$. A commonly used order-of-magnitude estimate for the current density in the scrape-off layer is given by the ion saturation current density, $j_\phi \sim j_{\text{sat}} = en_e c_s$, where $c_s = \sqrt{T_e/m_i}$ is the ion sound speed.

Quasi-neutrality is assumed, and the net current is expected to be mainly carried by electrons streaming along the field lines. This leads to a rough estimate of the toroidal current density $j_\phi \sim -5 \times 10^4$ A/m^2, where the negative sign indicates that the current is dominated by electron motion.

To estimate the magnetic field generated near the divertor leg surface, the scrape-off layer current is approximated as a planar current sheet of finite thickness λ_{SOL}. This parameter is generally estimated to be on the order of $1[cm]$[39]. In this approximation, the vertical component of the magnetic field in the vicinity of the sheet is given by

$$B_Z \sim \frac{\mu_0 J}{2} \sim 0.3 \text{ [mT]}, \quad \text{with} \quad J = \lambda_{\text{SOL}}\, j_\phi, \tag{34}$$

where J [A/m] is the surface current density. This model provides a deliberately upper-bound, order-of-magnitude estimate of the vertical magnetic field generated in the vicinity of the divertor manifold, as the leg is primarily vertical in the tentacle structures region. Owing to the assumption of an infinitely extended current sheet, spatial variations and finite-geometry effects are neglected.

As shown in sec.7.4, the estimated required δB^Z was in the range 1.3-2.6$[mT]$. It would require a surface current density at least 5 times higher, which seems unrealistic. These results are not conclusive regarding the SOP current's ability to induce a significant vertical magnetic field to explain the discrepancy in the average distance between the legs. The model of divertor current contains a lot of approximation. The decrease in electron density along the divertor leg and the current in the tentacle region is not taken into account.

Moreover, the current along leg is not possible to measure directly and can only be approximated roughly. It is possible that in the present shots, the current is significantly lower than the lower estimation made in this section. It could also be possible that there is larger λ_{SOP} accounting for the tentacle region.

The order of magnitude of the vertical magnetic field coming from an estimated SOP current, which is a high estimation, was too low to have an effect comparable to the discrepancies in sec.7.4. In the next chapter, adaptations are made on the LIUQE reconstruction code. The goal is to see if changing the axisymmetric reconstruction equilibrium has an effect on the manfiold shapes.

8.4 LIUQE adaptation

Here, different adaptations will be applied to LIUQE to see the effects on the turnstile lobes. The RAPTOR-LIUQE coupling equilibrium reconstruction will be used to see if it affects the manifolds. Subsequently, an increase and decrease in plasma current and magnetic probes will be implemented in LIUQE, and the equilibrium reconstruction will be recomputed and utilized as the new axisymmetric equilibrium in the `pyoculus` simulation.

The LIUQE code solves the Grad-Shafranov equilibrium equations by utilizing various measurements of the fields and current in external probes. However, the code lacks internal measurements, making the computation of the internal pressure and current density difficult to resolve. The turnstile lobe patterns are likely to be dependent on the internal current profile, which affects the q-95 value. This is the q-factor value of the surface that encloses 95% of the poloidal flux. In the reconstruction of LIUQE, the RAPTOR code can be coupled with it[40]. This code is a transport code that provides information about the current density and the pressure profile. A new reconstruction equilibrium ψ was given by S.Van Mulders, which employs the RAPTOR-LIUQE coupling. The simulations are subsequently conducted using this new axisymmetric reconstruction equilibrium. This is done on shot 77062

In fig.32a, a plot of the manifolds computed with the new reconstruction equilibrium and the old one is displayed. It shifted the turnstile lobe and the separatrix to the left. Even if the effect is subtle, it has still impacted the result. The fixed-points didn't move significantly, but the manifolds in the divertor region did move. The stable manifold on the LFS has moved to the right, and the divertor's manifold to the left.

The other adaptations that can be applied to the LIUQE reconstruction code is an artificial increase and decrease in the plasma current I_p and in the magnetic probes b_m. The equilibrium can be recomputed with these new values, and the ψ is used as input for the manifold simulations. Two different LIUQE reconstruction adaptations are presented to evaluate their effects on the magnetic tangles. These processes are neither rigorous nor consistent; however, it is conducted as exploratory measures to identify potential sources that affect the magnetic tangle patterns. A first equilibrium is computed, values are adjusted, and the new equilibrium is recomputed. The ψ function is then used as an axisymmetric equilibrium for the `pyoculus` solvers.

The first adjustment is a 5% increase in the plasma current $I_p^{new} = 1.05 \cdot I_p$ and in the magnetic probe measurement $\mathbf{b_m^{new}} = 1.05 \cdot \mathbf{b_m}$. The magnetic probes are also increased, as elevating the plasma current raises the poloidal magnetic field, which is captured by the magnetic probes. The second is a decrease of 5% of $I_p^{new} = 0.95 \cdot I_p$ and $\mathbf{b_m^{new}} = 0.95 \cdot \mathbf{b_m}$.

For the increase shown in fig.32b, the tentacles and the divertor's manifold have shifted to the left. The stable manifold on the LFS has shifted to the right. The fixed-points have also moved. It seems that the spacing between the manifolds has increased; however, this could be due to the shift to the left.

Conversely, for a decrease, shown in fig.32c, this is reversed. The tentacles and the divertor's manifold have shifted to the right, and they now connect to the LFS. There is an overlap between the divertor's manifold and a stable manifold directed toward the LFS. The fixed-points are also shifted. There seems to be a minor effect in reducing the distance between the manifolds; however, due to the significant shift, this is difficult to quantify.

Changing the plasma current significantly affects the magnetic field topology, making its effect on the lobe spacing difficult to estimate. However, even for substantial increases or decreases of the plasma current, the observed changes in lobe spacing appear to be mainly caused by a spatial shift of the manifolds. Nevertheless, the non-consistent approach adopted in these two LIUQE adaptations does not allow any conclusive statement to be drawn.

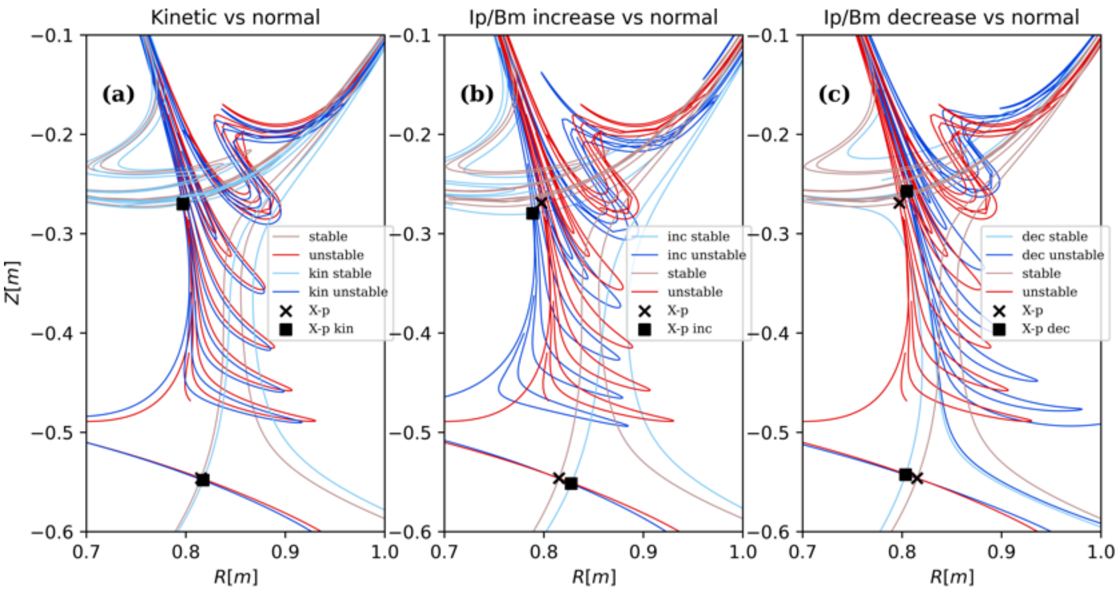


Figure 32: Manifold plots with error-field perturbations combined with three different axisymmetric equilibrium reconstructions, from LIQUE adaptations.

9 Conclusion

During this work, strong evidences was found for the presence of error-field-induced magnetic tangles in TCV. Observations from the MANTIS diagnostic showed specific emissivity patterns in certain plasma configurations, in which the magnetic tangle structures exhibited notable similarities.

Using non-calibrated tomographic inversion, two-dimensional poloidal emissivity profiles were reconstructed. These reconstructions were used as an experimental diagnostic to study the emissivity distribution in the divertor region, providing indirect information on the plasma density distribution.

The Poincaré map was then introduced in order to understand its principles and to use it for simulating magnetic field line trajectories. Adaptations were made to the `pyoculus` package to perform numerical simulations for TCV configurations. The effects of adding non-axisymmetric magnetic perturbations were discussed, with a focus on the turnstile transport process arising from the presence of magnetic tangles.

To compute the error fields associated with non-axisymmetric displacements of the poloidal field (PF) coils, a fitting method previously developed for TCV was described and applied to specific shots, such as back-off shots and strayshots. The resulting fits were compared, and the solution most consistent with the Jellyfish discharge dates and associated uncertainties was selected. Among the PF coil sets, the E-coils were found to exhibit the largest displacements, although these results were also associated with the largest uncertainties.

The equilibrium reconstruction code LIUQE was briefly introduced, as it provides the axisymmetric magnetic field data on a grid used as input for the Poincaré map simulations performed with `pyoculus`. The magnetic field perturbation resulting from the non-axisymmetric displacements of the PF coils were then described and numerically added to the axisymmetric magnetic configuration. Perturbed Poincaré maps were computed for a Jellyfish configuration, and turnstile lobes already appeared with spatial patterns that are qualitatively consistent with the structures observed in the MANTIS diagnostic.

The manifolds computed from the perturbed magnetic fields were overlaid on the MANTIS tomographic reconstruction images to enable a direct comparison between simulations and experimental observations. A qualitative visual analysis of the pattern shapes provided strong evidence of the correlation between the two, although small discrepancies were observed. Some of these discrepancies, such as the number and spatial location of the tentacle structures in specific discharges, were attributed to the use of the C-III spectral line. The combined effects of tomographic inversion uncertainties and the emission properties of the C-III line initially suggested the presence of an additional leg. However, this interpretation was ruled out by comparisons with helium line measurements and raw diagnostic data. Using alternative spectral lines led to a significantly improved spatial correlation. For further analysis, calibrated MANTIS data or Thomson scattering measurements could be used to obtain temperature and density information, allowing for a more accurate interpretation of spectral line emission, in particular for the C-III line.

The spatial and pattern shape correlations observed for shots 77021 and 75979, together with the absence of elongated turnstile lobes in non-tentacled shots, provide strong evidence that the tentacle structures are associated with turnstile lobes. To further support this interpretation, additional analyses on other shots and divertor configurations could be performed, as only five shots were considered in this study. Some discrepancies remain unexplained, such as the length of the tentacles and their spatial periodicity. Since several physical processes are not included in the simulations, such as perpendicular transport, a perfect agreement is not expected. Considering these limitations, the accuracy of the predicted tentacle locations remains surprising. The spatial periodicity was estimated using a one-dimensional line projection and was correlated with a possible constant uncertainty in the poloidal magnetic field.

The effects of specific E-coil configurations on the turnstile lobe patterns were discussed. Coil tilts and center shifts produced comparable effects, while their combination led to either constructive or destructive interference, depending on their relative locations. The displacement of the E4 coil was found to have a strong impact on the turnstile lobe patterns, due to its location and current intensity.

Finally, the poloidal magnetic field generated by currents flowing along the divertor manifolds was roughly estimated, and several adaptations of the LIUQE code were tested. However, these attempts did not lead to conclusive results.

In this work, a method for computing error-field-induced magnetic perturbations was described and implemented for application to TCV. This approach may provide a useful basis for future studies investigating other effects associated with error-field-induced non-axisymmetric magnetic perturbations.

A Additional Material

This appendix contains supplementary information that supports the main text.

A.1 Incorporating magnetic probes into the displacements fitting

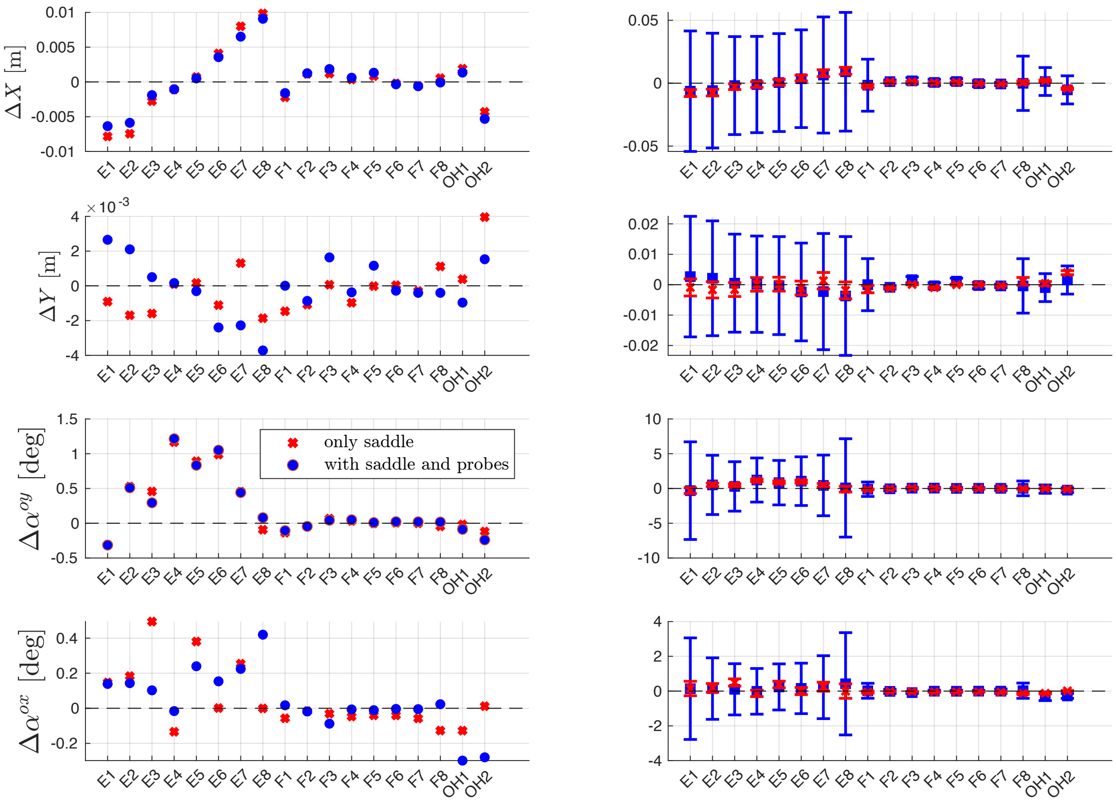


Figure 33: PF coil displacements and tilts computed with the fitting of eq.(36) and eq.(15). Two model are used, with or without the addition of magnetic probes.

Theoretically, displacements ($\mathbf{\Delta X}$, $\mathbf{\Delta Y}$,$\mathbf{\Delta \alpha_x}$, $\mathbf{\Delta \alpha_y}$) cannot be computed using a single magnetic probe sector [7]. However, since there are two sectors, they can also be used to infer the displacements. The calculation is similar to that for $\Delta\psi_s$; the only difference is that the probe measures the fields locally. The term $\cos(\theta_m)$ can be used instead of the trigonometric average. They are given by the following expressions,

$$\Delta_{in}\mathbf{b_m} = \mathbf{b_m} - \underline{\underline{\mathbf{B_{mc}}}}\mathbf{I_c} \ , \tag{35}$$

$$\begin{aligned} \Delta\mathbf{b_{m,i}} = &-\frac{\partial \underline{\underline{\underline{\mathbf{B}_{mc}}}}}{\partial r_m}\cos(\theta_m)\mathbf{I_{cc,i}}\mathbf{\Delta X_c} - \frac{\partial \underline{\underline{\underline{\mathbf{B}_{mc}}}}}{\partial r_m}\sin(\theta_m)\mathbf{I_{cc,i}}\mathbf{\Delta Y_c} \\ &-\frac{\partial \underline{\underline{\underline{\mathbf{B}_{mc}}}}}{\partial z_m}\cos(\theta_m)\mathbf{I_{cc,i}}\mathbf{\Delta_t Z_c^x} - \frac{\partial \underline{\underline{\underline{\mathbf{B}_{mc}}}}}{\partial z_m}\sin(\theta_m)\mathbf{I_{cc,i}}\mathbf{\Delta_t Z_c^y} \equiv \underline{\underline{\mathbf{A}}}^{b_m}\mathbf{x} \ . \end{aligned} \tag{36}$$

The term $\underline{\underline{\mathbf{B_{mc}}}}^{76\times 18}$ is the magnetic inductance between a probe and a coil. The weight is expressed as $\delta b = 0.5[mT]$[41]. This time, the matrix $\mathbf{A} = (\underline{\underline{\mathbf{A}^{\psi}}}, \underline{\underline{\mathbf{A}^{b_m}}})$ and the inputs are $\mathbf{b} = (\Delta_{in}\psi_{\mathbf{s}}, \Delta_{in}\mathbf{b_m})$.

The result of the displacement calculation fitting with backoff 78000 is shown in fig. 33. The blue squares represent the results obtained using both diagnostics, while the red crosses indicate the results from only the saddle loop measurements $(\underline{\underline{\mathbf{A}^{\psi}}}); (\Delta_{in}\psi_{\mathbf{s}})$. The matches are suitable for ΔX and $\Delta\alpha^{oy}$. For the other two, the results are more sparse. Their spatial resolutions are toroidally limited, consisting of only two sectors that are radially opposed. This limitation could be attributed to their proximity to the Y axis. Their contribution is more important for the $\sin(\theta)$ dependent value, such as ΔY and $\Delta\alpha^{ox}$. Adding the probes seems to introduce a considerable amount of uncertainty into the calculations. As illustrated in fig.33, for ΔX, the error bar ranges from 5 to 10 cm for blue compared to 0.5 to 1 cm for red. This indicates that the probes are not effective diagnostics for measuring differences between probes in sector 11 and sector 3. Their calibration seem to be not precise enough to have a reliable sensitivity for measuring differences in the mT order. Since they measure $\frac{dB}{dt}$, the value of the field B is integrated, making uncertainty propagation even more important. These are the reasons why PF coil displacements are computed using only the saddle loop diagnostic.

A.2 Resonance zone area calculation

In this chapter, the calculation of the resonance area of a magnetic island chain using Meiss's action principle was implemented. This is described in this section. First, a quick description of a resonance zone of a magnetic island is provided. Then a quick description of the action principle and finally, plots and convergences between other area formulas and Meiss's[19] action formula.

The process of turnstile transport is also responsible for transport between the regions bounded by the magnetic island chain of an (m, n) orbit. An example is shown in fig.34. The shaded region represents the resonance zone of the (1,3) standard map. In this instance, the region R is defined as the area inside the curve formed by 2 stable segments and 2 unstable segments. $\partial R = S^1 + S^2 + U^1 + U^2$. The top segment is formed by $\partial S^1 = m_1^+ - H_r$ and $\partial U^1 = H_l - m_1^+$. The bottom segment is formed by $\partial S^1 = m_1^- - H_r$ and $\partial U^1 = H_r - m_1^-$. H_l and H_r are the two fixed points located on the left and right sides of the resonance region. m and h are heteroclinic orbits, as they are on manifolds originating from distinct fixed points. There are two entry sets $I = I_u \cup I_l$ whose pre-image is depicted in violet. The exit set is defined by $E = E_u \cup E_l$. The turnstile lobes represent the image/ pre-image of E and I.

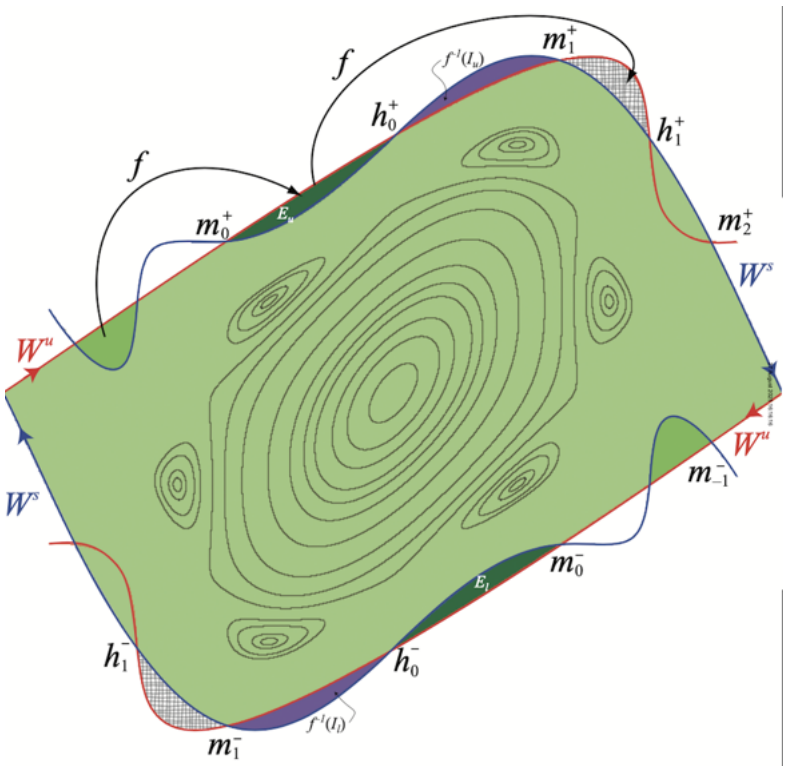


Figure 34: Resonance zone of a (1,3) orbit of the standard map. The resonance region is in green. Taken from Meiss (2015)[19].

A.2.1 Computing resonance area with Meiss's action principle

For a diffeomorphism, volume are computed using a differential form $\beta = \rho(x)dx^1 \times ... \times dx^n$ and the volume measure $\mu(R) = \int_R \beta$ for a measurable set R. The integration of magnetic field lines in a tokamak defines a Poincaré map f associated with one toroidal turn. Owing to the divergence-free property of the magnetic field, $\nabla \cdot \mathbf{B} = 0$, this map preserves the magnetic flux and therefore the induced area form β on the Poincaré section,

$$f^*\beta = \beta. \tag{37}$$

Introducing locally a one-form α such that $\beta = d\alpha$, the quantity $f^*\alpha - \alpha$ is a closed one-form, $d\,(f^*\alpha - \alpha) = 0$. Assuming a simply connected domain of the Poincaré section, and no net magnetic flux variation across the section, this closed form is exact. Consequently, there exists a scalar function L, defined locally on the Poincaré section, such that

$$f^*\alpha - \alpha = dL. \tag{38}$$

The function L can be interpreted as a generating function associated with the area-preserving Poincaré map of magnetic field lines. These definitions become clearer when they are related to the magnetic field line formalism. The two-form β can be identified with the infinitesimal magnetic flux through a surface element of the Poincaré section, $\beta = \mathbf{B}\cdot d\mathbf{S}$, so that the measure $\mu(R)$ associated with a region R of the section corresponds to the magnetic flux $\Phi(R)$ through this region. Locally on the Poincaré section, a one-form α can be introduced such that $\beta = d\alpha$, where α may be identified with the magnetic vector potential $\mathbf{A}$ in a given gauge. Under this local identification, Stokes' theorem yields $\Phi(R) = \int_R \beta = \int_{\partial R} \alpha \equiv \int_R \mathbf{B} \cdot d\mathbf{S} = \int_{\partial R} \mathbf{A} \cdot d\mathbf{l}$. The computation of the turnstile lobe flux can then be expressed as a sum of Lagrangian contributions using eq.(38), where L is a scalar function defined locally on the Poincaré section. For the magnetic field line Poincaré map, the Lagrangian associated with a point p is defined as the line integral of the vector potential along the magnetic field line,

$$L(p) = \int_{-2\pi m}^{0} \mathbf{A} \cdot d\boldsymbol{\ell}, \tag{39}$$

where m denotes the number of iterations of the corresponding Poincaré map f^m. For the magnetic flux contained in a turnstile lobe formed by a broken separatrix around the magnetic axis, and delimited by a pair of homoclinic points h_t and m_t, the resulting expression is given by

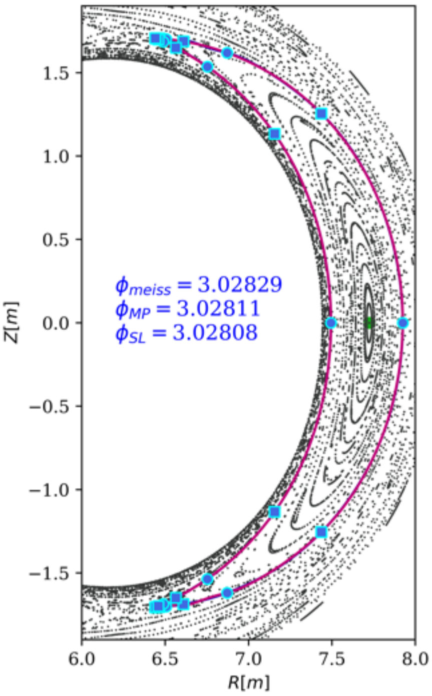
$$\phi_{\text{turnstile}} = \sum_{t=-\infty}^{\infty} \left[L(m_t) - L(h_t)\right], \tag{40}$$

as derived in ref [19].

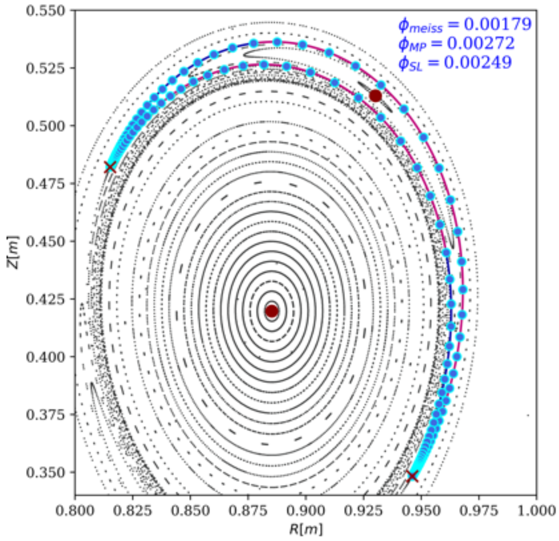
For a resonance zone, the region is bounded by four points: two hyperbolic fixed points H_l and H_r, and two heteroclinic points m_t^+ and m_t^-, as shown in fig.34. The corresponding flux, was rigorously derived in a R.MacKay's work (1987)[42] and is given by

$$\phi_{\text{reso}} = m \sum_{t=-\infty}^{\infty} \left[L(m_t^+) - L(H_l) + L(m_t^-) - L(H_r)\right], \tag{41}$$

where m is the number of islands in the chain. The Lagrangian integration therefore involves the same integer m. Eq.(41) is implemented in the `pyoculus` code and is tested on an analytical example and on a TCV discharges. In fig.35a, the resonance zone of an analytical tokamak configuration with a Maxwell-Boltzmann field perturbation is display and in fig.35b, resonance zone of a m=2 island in an error-field TCV doublet configuration. The resonance area is compared to an improved shoelace and midpoint formula. For the shoelace, the contour of the resonance zone is taken, using the contour formed by 4 manifolds segment intersecting at 2 heteroclinics. For the midpoint rule, the magnetic field $\bar{B_{in}}$ averaged over a number of inside point is used. For the TCV example, the three values show a significative difference. This could be caused by the large resolution of the grid $\Delta R \simeq 0.02[m]$ which is on the same order as the thickness of the island. The midpoint rule is supposed to be more precise as it wasn't integrated like the vector potential $\mathbf{A}$. The Meiss formula is less practical for this example, as a high number of heteroclinics lie on the manifolds.



(a) Analytical tokamak with MB (3,2) perturbation.



(b) Resonance zone area for a doublet configuration in TCV.

Figure 35: Resonance zone calculation using Meiss action principle. It converges to the improved shoelace and midpoint formulas

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